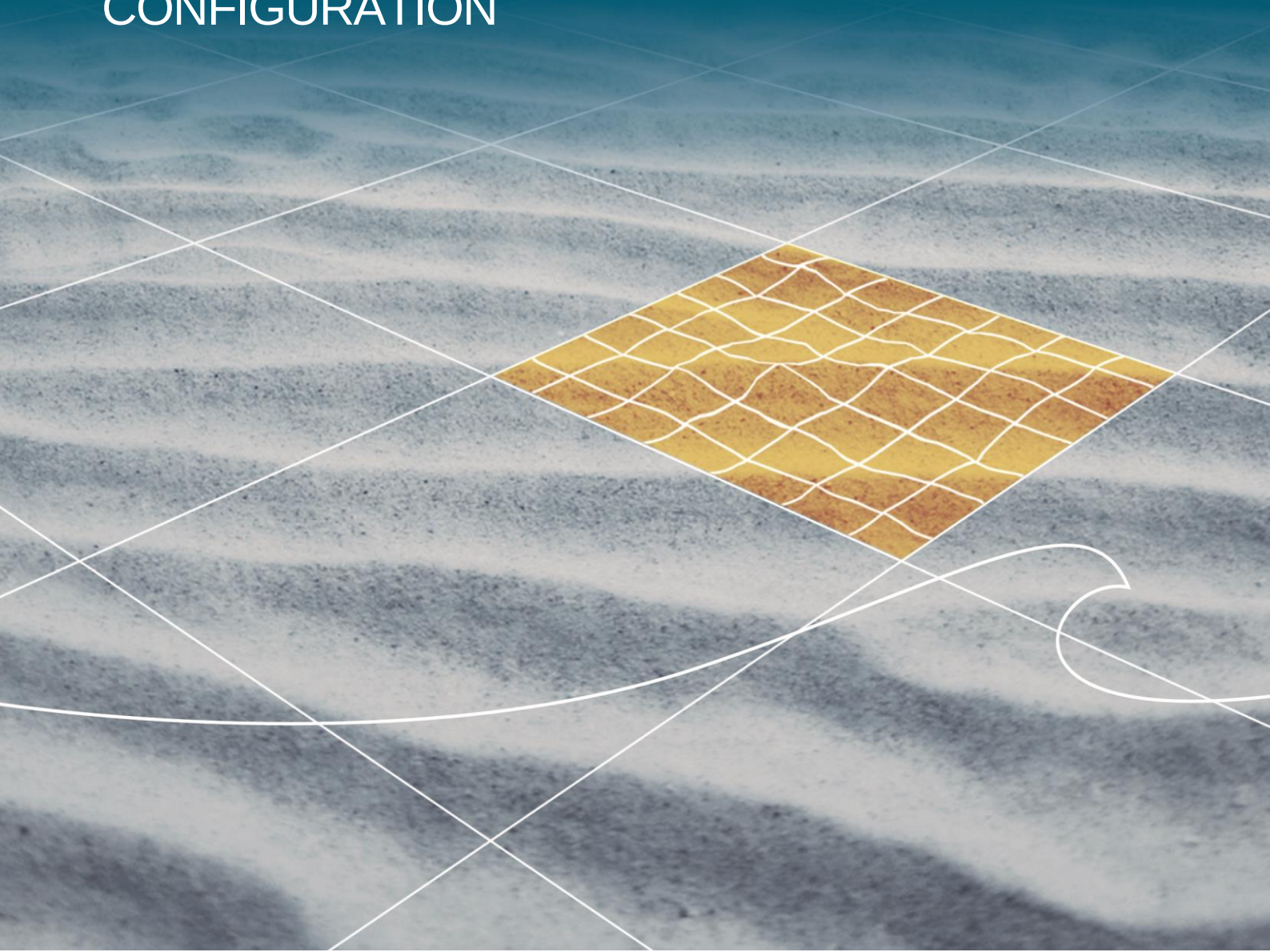
EIVA
MARINE SURVEY SOLUTIONS

NAVIPAC 4.5

CONFIGURATION



Contents

1 The main window 4

1.1 Vehicles view 6

1.2 3D visualisation 7

1.3 Tree view 9

1.4 Properties view 11

1.5 Controls 13

2 The main menu 15

3 Instruments 18

3.1 Edit instrument properties 18

3.2 Edit I/O settings 21

3.3 Edit special settings 23

3.4 Remove instrument 23

3.5 Add instrument 23

3.6 Sort instrument list 27

3.7 Dynamic positioning 27

4 Geodesy 29

4.1 Projection and ellipsoid 30

4.1.1 EPSG 32

4.2 Datum shift 33

4.2.1 Helmert 7 35

4.2.2 Mod. Bursa-Wolf 35

4.2.3 NADCON US 35

4.3 ITRF 36

4.4 Height reduction 38

5 Project settings 40

5.1 NaviPac 41

5.2 Warm start 43

5.3 GPS 44

5.4 NaviPac filters 46
5.5 Underwater and remote positioning 48
5.6 Advanced 49
5.7 Survey parameters 51
5.8 Rig move 52
5.9 Telemetry setting 53
5.10 NaviPac eventing and triggering 54
5.11 NaviScan 56
6 Navigation 58
7 References 60
8 Figures 61

1 The main window

The NaviPac configuration module **NaviPac Configuration** is the base program for setting up an entire NaviPac operation.

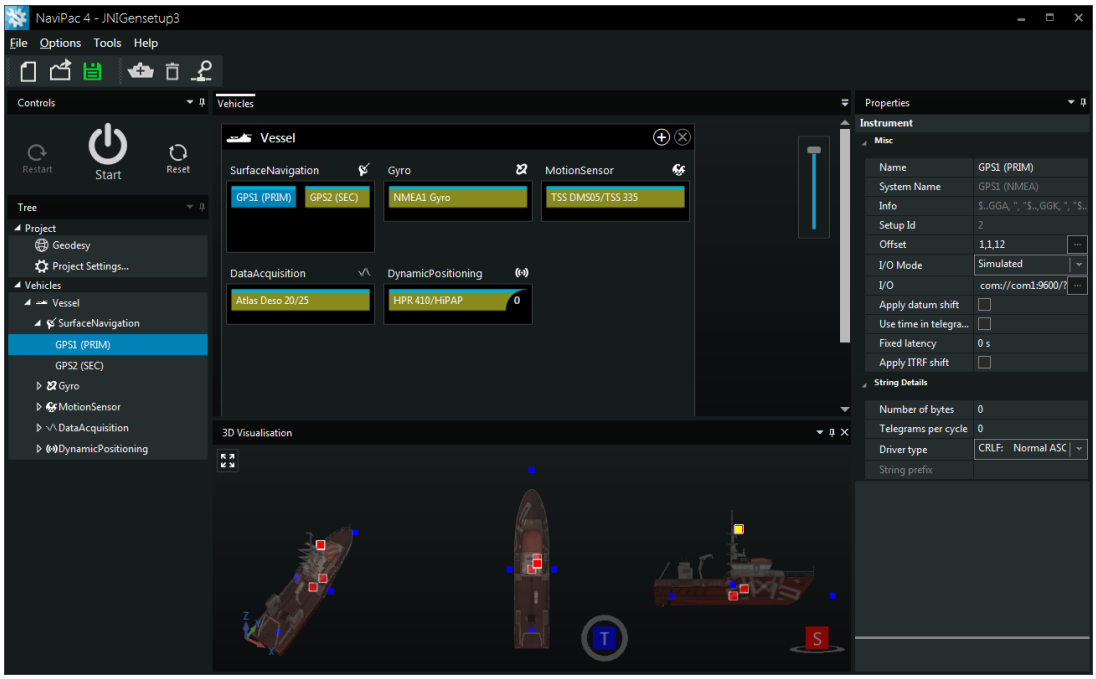


Figure 1 NaviPac Configuration main window

The main window shown above opens when NaviPac Configuration is started. The program automatically loads the setup file you were using last time.

Note: You may edit any setup file with NaviPac Configuration; name and location can be chosen freely.

The main window can be resized to make it larger and smaller in size. The following sections walk through the program as it is defined by default – but please note that the operator can move and place windows as they choose.

NaviPac Configuration is divided into a series of views and controls showing various parts of the setup, but all parts are linked. This means that selecting an item will automatically select/show the same item in the other views.

Before using the software, there are some basic terms that need to be pointed out.

- **Vehicle**
  Any object that has been positioned by NaviPac is called a **vehicle** – so the main vessel, ROVs, ROTV, AUV and remote vessels are all determined as vehicles.
- **POI** (Point of interest)
  A fixed point on a vehicle is called a POI. (POI were previously called **User offsets**.) A POI can be used in data outputs or for assigning items to a location, among other uses.
- **Instrument**
  The term **instrument** covers both physical instruments (like GPS) and general data suppliers (like AIS) and data sent out of NaviPac. Instruments will be grouped in sections depending of the type of instrument, which types include:
  - **Surface Navigation**
    Positioning of main vessel using GPS, tracking system or similar
  - **Gyro**
    Gyro, compass and similar
  - **Motion Sensor**
    Roll, pitch and heave sensors
  - **Data Acquisition**
    Input of acquisition sensor data such as echo sounder, magnetometer, pressure-, pipe- and cable tracker etc
  - **Dynamic Positioning**
    Positioning of remote and under water vehicles such as USBL, LBL, and remote GPS
  - **Doppler Log**
    Speed logs used for vehicle positioning
  - **Data Output**
    Data output to external devices
  - **Misc**
    Anything that doesn't fit the other categories, such as GPS time sync or radar input

1.1 Vehicles view

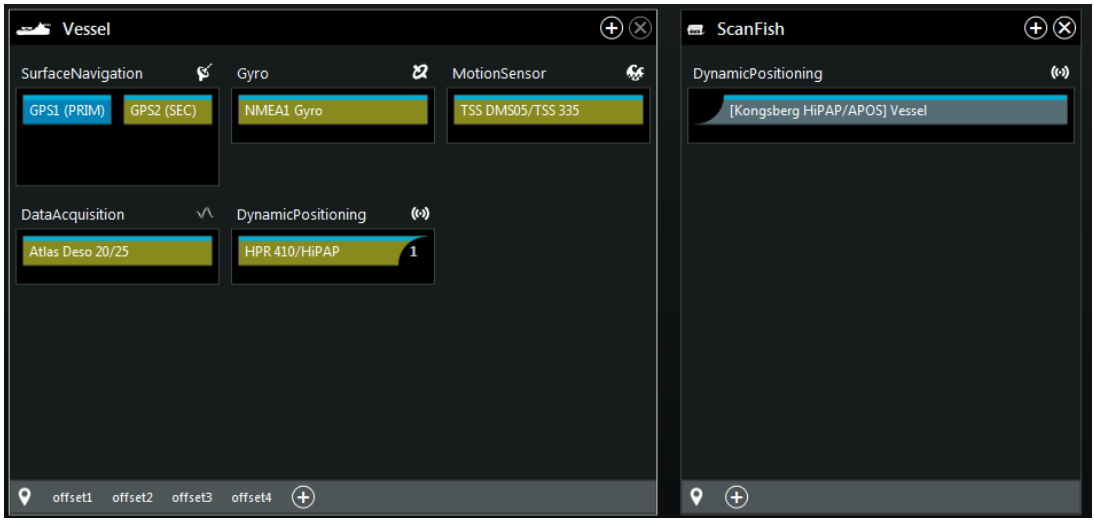


Figure 2 Vehicle view illustration main vessel and ROTV

The **Vehicle** view gives an overview, with a section per vehicle and subsections inside each with instruments, and can be named freely. The lowest part of the view gives attached POIs (if any). A series of functions can be activated from this view:

- **Select a vehicle**
  Click on the title bar in the vehicle box. The vehicle will simultaneously be selected in **Tree** view, **Properties** view and **3D** view.
- **Select an instrument**
  Click on the desired title bar inside the vehicle box (in the figure above these titles include SurfaceNavigation, Gyro, MotionSensor, DataAcquisition, and DynamicPositioning). The instrument will simultaneously be selected in **Tree** view, **Properties** view and **3D** view.
- **Select POI**
  Click the POI in the bottom bar of the vehicle box. The POI will simultaneously be selected in the **Tree** view, **Properties** view and **3D** view
- **Add instrument** (instrument will be selected after addition)
  - Click the **+** icon at the vehicle title bar
  - Right-click inside vehicle box
  - Right-click inside instrument group box (selection is filtered to same type)
- **Add POI** (POI will be selected after addition)
  Click the **+** icon at the bottom bar
- **Remove instrument**
  Right-click on the instrument box and select remove
- **Remove POI**
  Right-click at the POI in the bottom bar and select remove

- **Remove vehicle**
  Click the **X** button at the upper right part of the vehicle bar
- **Move instrument from one vehicle to another**
  Drag-and-drop instrument box (click and move)
- **Copy instrument** (both at same vehicle and to another)
  Drag-and-drop instrument box (Ctrl + click and move)
- **Copy vehicle**
  Drag-and-drop vehicle box (Ctrl + click and move)

Instrument boxes will be coloured by **Input mode** (**Simulated**/**On**/**Off**) and will show linked information (eg between vessel USBL system and ROV transponder).

1.2 3D visualisation

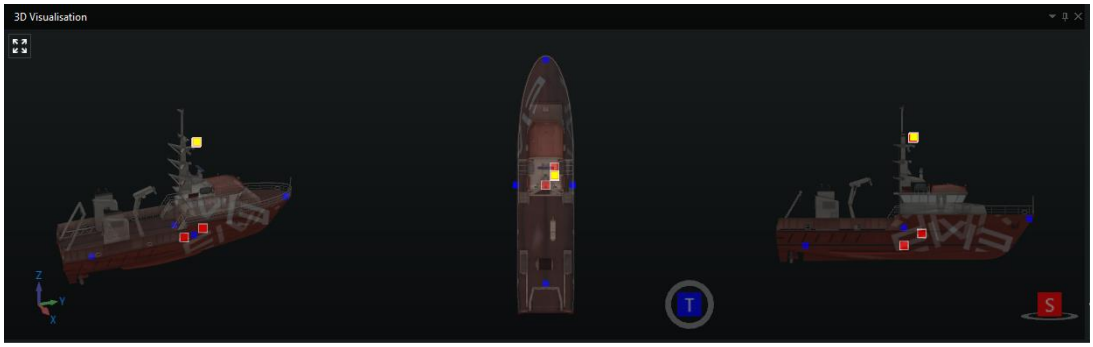


Figure 3 3D visualisation of vehicle and offsets

The **3D Visualisation** view shows the selected vehicle in a 3D environment in three views – they can be zoomed into and out from individually by using the mouse wheel. The views are as follows, from left to right in the above figure:

- **Free 3D**
  Rotate by clicking and holding down right mouse button
- **Top view**
- **Side view**

To use the **3D Visualisation** view optimally, you need to assign a 3D drawing to the vehicle. This is done by selecting the vehicle and defining the drawing in the **Properties** view:

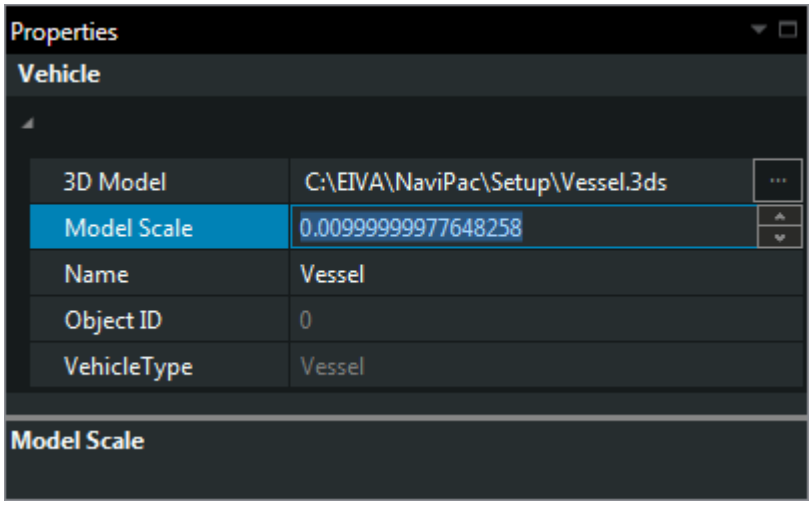


Figure 4 Define vehicle drawing and scale

The drawing must be in the data type 3DS (defined by, among others, Autodesk 3ds Max) as we also use this in Helmsman's Display and NaviModel. The files used in the configuration will work best if they are drawn in semi-transparent mode.

All instruments and POIs will be drawn in the images so you easy can evaluate the locations because they are represented by boxes, keyed as follows:

- **Red box**
  All instruments are represented by a small red box
- **Yellow box**
  The selected instrument is represented by a yellow box
- **Blue box**
  All POIs are represented by blue boxes

1.3 Tree view

The left pane of the main window consists of the **Tree** view (a listing and control tool) for the entire project, as seen in the image below

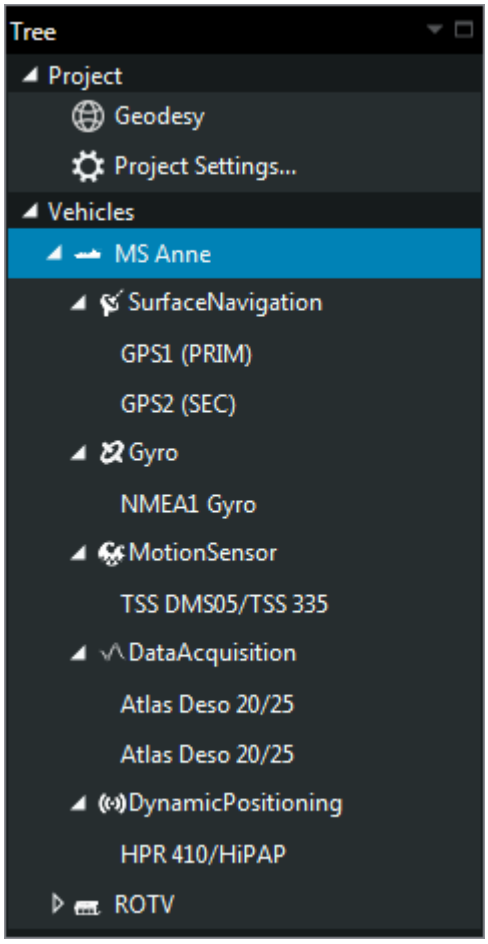


Figure 5 NaviPac Project tree

Selecting an item will automatically select the corresponding item in the **Properties** view (for definition) and in the other views for display and control

- **Project**
  Contains general project-related information
  - **Geodesy**
    Definition of projection, ellipsoid, ITRF and height reference. See chapter on geodesy later in this manual for further details.
  - **Project Settings**
    Definition of project-specific items (formerly known as **Global Parameters**) See chapter on project settings later in this manual for further details.

- **Vehicles**
  List of all vehicles in the configuration. The **Vehicles** view is linked to the vehicle box and the **3D Visualisation** view. You may add instruments to a vehicle by right-clicking the vehicle or the instrument item:
  - **Surface navigation**
    Any instrument used for surface navigation such as GPS or tracking systems
  - **Gyro**
    Gyro and compass for vessel and dynamic objects
  - **Motion sensor**
    Roll/pitch heave sensors for vessel and dynamic objects
  - **Doppler log**
    Doppler velocity log for positioning improvement
  - **Dynamic positioning**
    Positioning of remote or underwater vehicles such as USBL, LBL, remote GPS, tracking systems, telemetry, etc
  - **Misc**
    Dedicated drivers for time, waypoints, targets, etc
  - **Data output**
    Output to external systems and instruments
  - **Data acquisition**
    Echo sounders, magnetometers, etc
- **Instrument**
  Edit a specific instrument by clicking on it; remove an instrument by right-clicking on the instrument.

1.4 Properties view

If an item is selected in the **Tree** view, properties for the selected item will open in the **Properties** view on the right side of the main window, allowing you to edit the item's properties (as shown for a gyro in the image below):

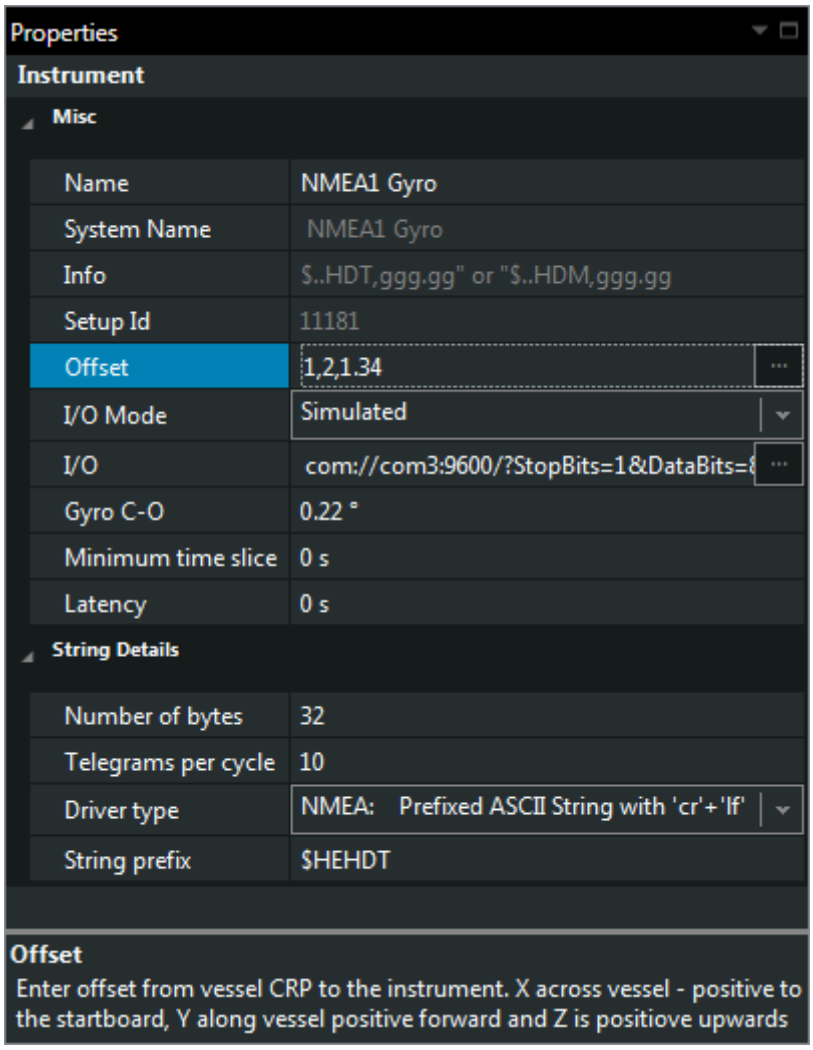


Figure 6 Instrument properties for a gyro

The various properties are described in more detail in the following section. Editing is based on a few basic controls.

- **Editable fields**
  Change texts, integers or floating-point values. Click on the field to edit the value. Changes are accepted once you click on a different field.

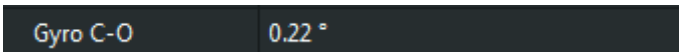


Figure 7 Editing a number or text in a field

- **File selection**
  Display the actual file. Click the **…** button to the right and a window will open which allows you select a file.

Figure 8 Selecting a file via browser function

- **Offsets**
  Special selection tool for offsets and POI selection. Click the **…** button to the right to select.

Figure 9 Offset selection

- **Lists**
  Select an item from a drop-down list. Click on the downward arrow button on the right side, and a drop-down list will appear with a checkmark shown to the left of the currently selected item. Changes are accepted as soon as a new selection is made.

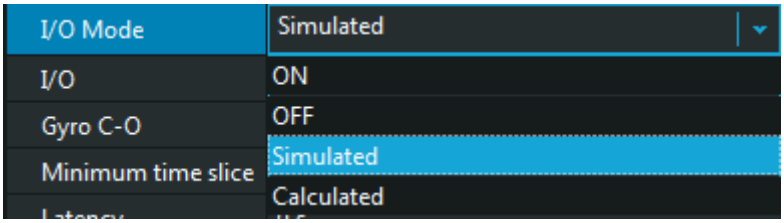


Figure 10 Selection from a list

- **Check boxes**
  Enabling or disabling of certain features. Click once inside the check box to select or clear the check box. Changes are accepted as soon as a new selection is made.

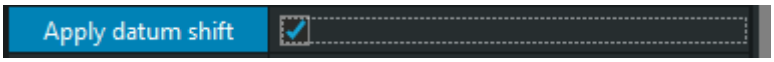


Figure 11 Using check boxes

- **I/O selection**
  Lists the actual I/O setting and allows changing via the built-in COM port tester, which is activated by clicking the button on the right side. **I/O selection** requires that NaviPac online navigation is stopped.

Figure 12 I/O selection

Depending on the configuration, some properties are read-only (that is, not editable). These property fields are greyed out.

1.5 Controls

NaviPac contains three important items for controlling the navigation process:

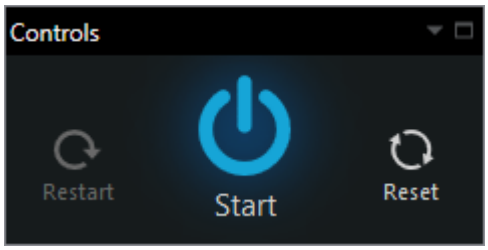


Figure 13 NaviPac control buttons

- **Restart**
  Perform a fast warm start of the navigation process. Unsaved changes in configuration (if any) will **not** be included.
- **Start**
  This is the general start (and stop) process. Perform a warm start if possible, or a manual start if a warm start is not possible. The colour of the icon tells if navigation is running or not.
- **Reset**
  Reset navigation status to ensure that a manual start is performed next time.

1.6 Toolbar

A small toolbar is available with the most commonly used tools, which are, from left to right:

Figure 14 NaviPac toolbar

- **Create a new configuration**
- **Open configuration**
- **Save configuration**
- **Create a new vehicle**
- **Delete selected item**
- **Display all instruments ordered by port** (the dialogue box seen in the figure below will appear when this is used)

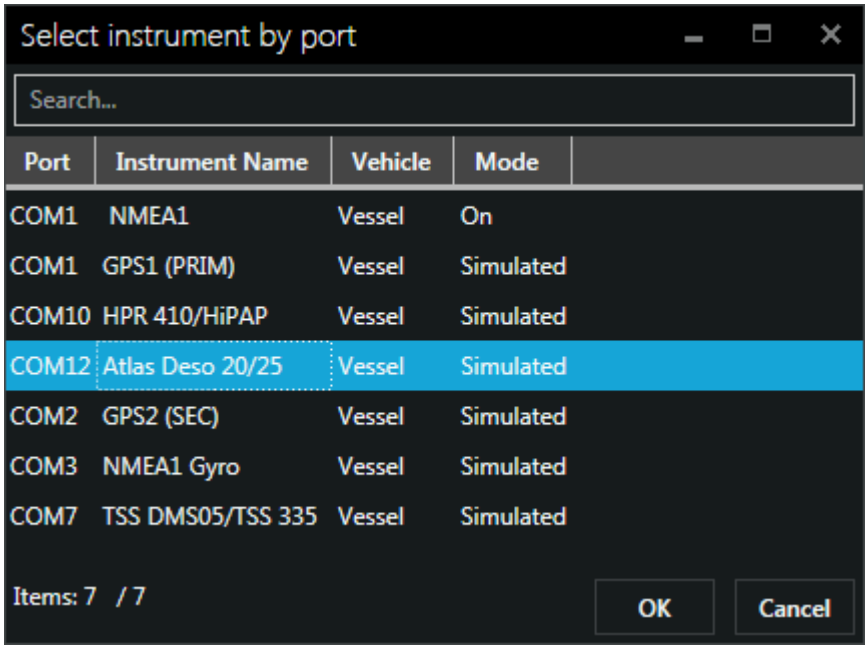


Figure 15 Select instrument by port

2 The main menu

The main menu gives access to a few general commands

- **File > New**
  Create a new empty configuration file.
- **File > Open**
  Open a pre-existing configuration file.
- **File > Recent**
  Open configuration from list of recent used files.
- **File > Save**
  Save the current configuration file.
- **File > Save as**
  Save the current setup file under a configuration name.
- **Options > Simulate all**
  Set sensors to simulated I/O Mode
- **Options > Confirm on Stop**
  Always prompt user before stopping navigation
- **Options-> Remoting**
  Enable Remote Supervision or take control of remote setup
- **Tools > GEOCalc**
  Open a small utility module for testing of the EIVA geodesy library etc.
- **Tools > Graphics Editor**
  Open a small utility module for graphical file import/export.
- **Tools > Project Manager**
  Opens the Project Manager backup and restore projects
- **Tools > Rig/Barge Setup**
  For Rigmove and tug management – define barge layout, fairleads and anchor spread
- **Help > Introduction**
  Small **Help** file with basic guidance to NaviPac Configuration
- **Help > Manual**
  Open this **Help** file (NaviPac Configuration).
- **Help > Data Formats**
  Open **Help** file with short listing of all supported instruments (NaviPac Sensor Interfacing).
- **Help > Geodesy**
  Open **Help** file with detailed information about EIVA geodesy (NaviPac Geodetic Control).
- **Help > Instrument Setup**
  Open **Help** file with special hints for instrument setup (NaviPac Instrument Parameters).
- **Help > User Defined Output Setup**
  Open Help file for User Defined Output setup.

- **Help > Trace Settings**
  Open a dialogue box for editing of a series of NaviPac Trace Settings

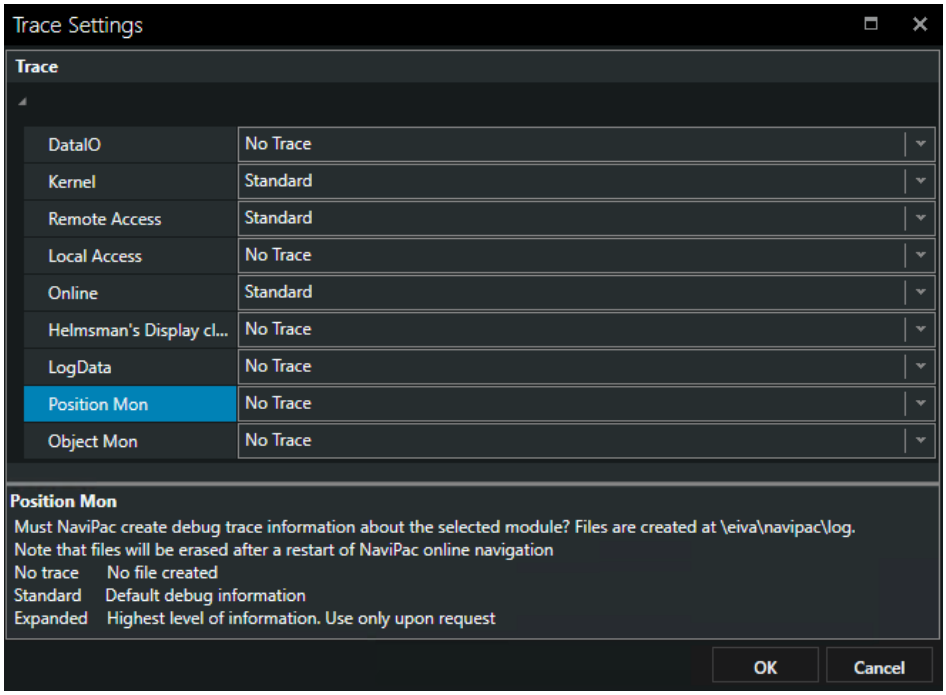


Figure 16 Trace Settings view

- **Help > Release Notes**
  Open **Help** file with latest release notes.

- **Help > License Activator**
  Automatic or manual activation of software license – common to all EIVA modules.

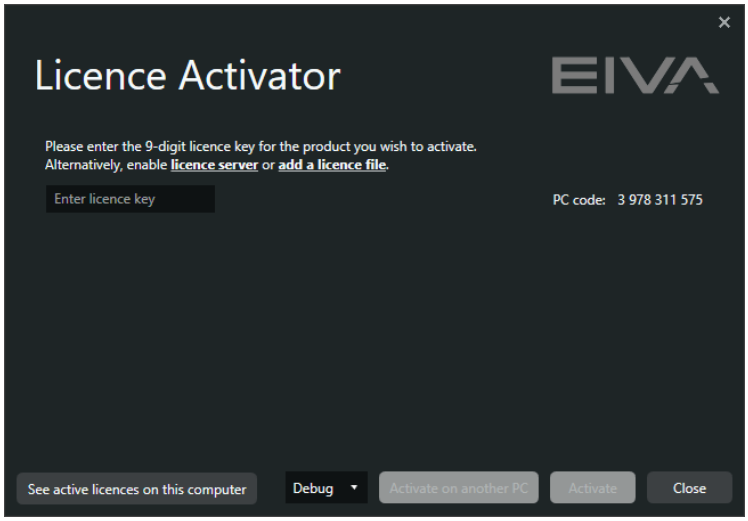


Figure 17 NaviSuite License activator

- **Help > About**
  Show current program version and dongle information.

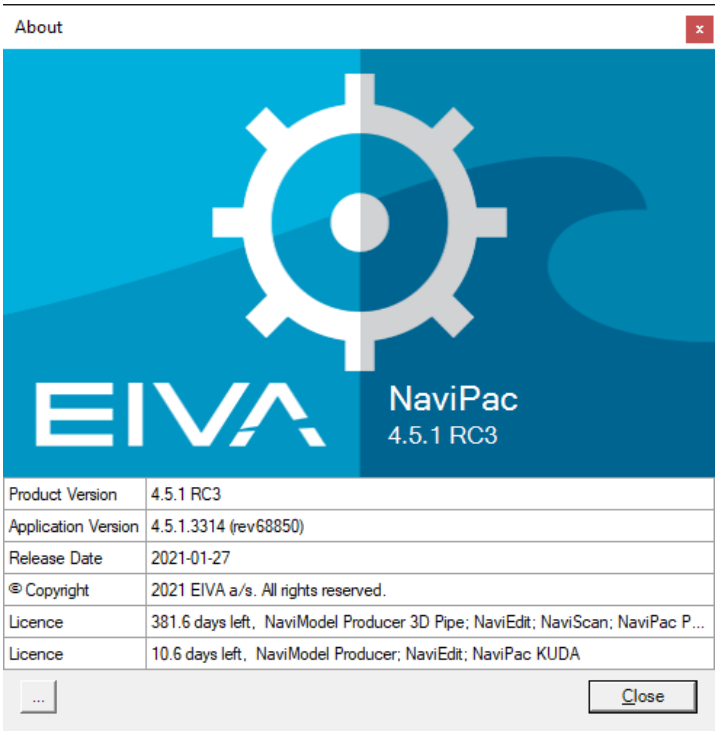


Figure 18 About EIVA NaviPac 4

3 Instruments

In NaviPac Configuration, the user may edit various properties for an instrument, edit I/O settings, edit special settings, remove instruments, add new instruments and edit the instrument order.

3.1 Edit instrument properties

To edit instrument general properties, you simply have to select the instrument from the **Tree** view or in the **Vehicles** view. The available properties will then appear in the **Properties** view.

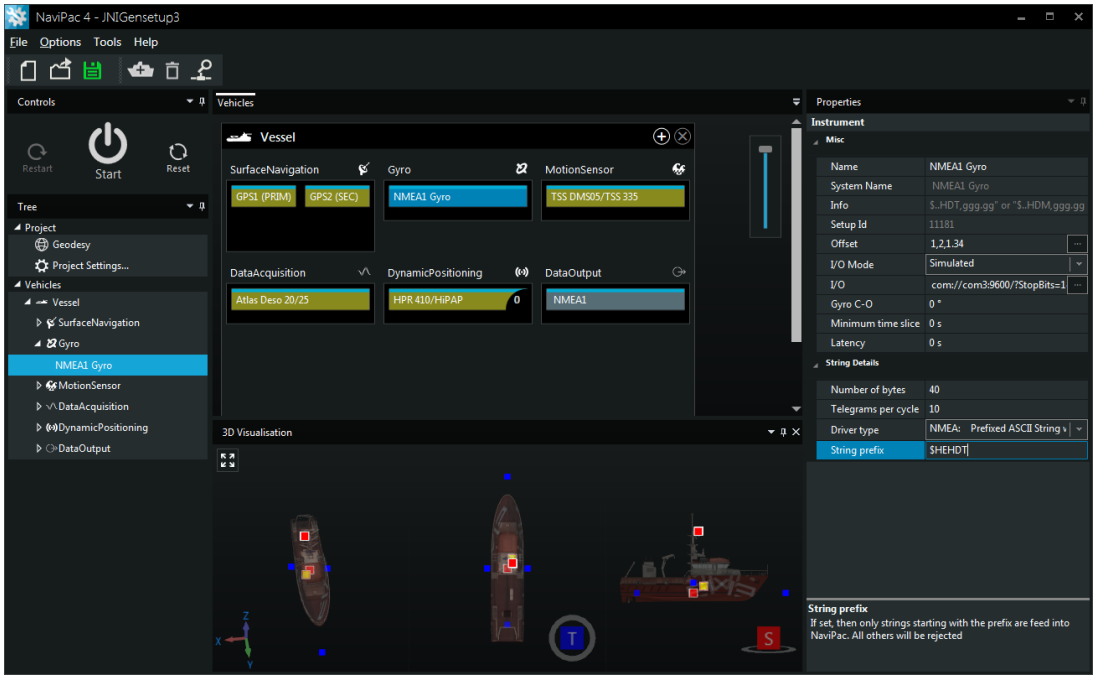


Figure 19 Edit instrument properties for a gyro

The **Properties** view will then consist of a series of general properties (common for the majority of all instruments) and perhaps some instrument group-specific properties. A list of instrument group-specific properties and properties unique to individual instrument types is available in the NaviPac Instrument Parameters **Help** file, which can be accessed under **Help > Instrument Setup**.

General properties

- **Id**
  Read-only. Displays unique NaviPac instrument type number.
- **Setup Id**
  Read-only. Displays NaviPac runtime ID.
- **Name**
  Allows for entering a user-defined name for the instrument.
- **System Name**
  Read-only. Displays NaviPac-defined instrument type name.
- **Info**
  Information about instrument data format.
- **Offset**
  Specify the location of the instrument. Defined from CRP to instrument:
  - **Offsets X**: Across vessel – positive to starboard
  - **Offsets Y**: Along vessel – positive to bow
  - **Offsets Z**: Height – positive up
- **I/O Mode**
  Select the input mode of the instrument:
  - **On**: Real interfaced instrument. (Not available in demo mode.)
  - **Off**: Temporarily disabled instrument.
  - **Simulated**: Instrument simulated by SIMDATA. (Useful for demonstrations and training.)
  - **Calculated**: Calculation based on vessel movements. (Only valid for gyro and Doppler log instruments.)
- **I/O**
  Edit I/O settings. Can also be accessed by right-clicking on the specific instrument in the tree view and selecting **Edit settings**.
- **String Details**
  For data input instruments (though not special input instruments), you may view and edit various properties for the specific driver. Please note that values specified for the following properties will be applied to all instruments of the same type (same **ID** number):
  - **String Prefix**: If the instrument outputs multiple telegrams of which NaviPac only uses one, this field allows you to define the string prefix. NaviPac will hereafter only use the telegrams that start with the defined string prefix (eg $HEHDT for NMEA gyro).
  - **Number of bytes**: NaviPac allocates a number of bytes (characters) for each incoming telegram. If this number is, for some reason, less than the actual instrument output (especially seen for user-defined input instruments), NaviPac truncates the telegram. You can enter a new, higher value here in order to see the actual instrument output with all characters included.

- **Telegrams per cycle**: NaviPac allocates space for a maximum number of packets per cycle. If this number is set too low, NaviPac will start discarding packets. You can enter a new, higher value here in order to allocate enough space for a higher number of packets.
- **Driver type**: NaviPac supports different input types, which each offer a unique way of recognising a complete telegram:
    - **CRLF**: The input is considered ASCII, and each string is terminated with two special characters (eg <CR><LF>). NaviPac uses the last character (<LF>) to detect that a packet is completed.
    - **CR**: Same as CRLF – except only with one special end character (eg <CR>).
    - **NMEA**: Same as CRLF – but NaviPac also uses a configurable string prefix (see above) to select between multiple telegrams.
    - **LENGTH**: NaviPac reads a chunk of data with fixed length – typically binary input. This may be combined with either a defined start character or end character.
    - **STX/ETX:** Packets with one special character at the start and one special character at the end. (These were mainly used many years ago with binary code <STX> and <ETX>.)
    - **XXXX**: Special telegrams that have been hardcoded by instrument number (eg RDI Workhorse Navigator DVL PD0 format and Raytheon Radar).
- **Termination char**: Read-only. If the driver expects a specific character at the end of the telegram, they can define it in this field. This field displays the ASCII code (eg '10' for <LF>) and is optional.
- **Start char**: Read-only. If the driver expects a specific character at the start of the string, they can define it in this field. This field displays the ASCII code (eg 68 for 'D') and is optional.
- **Fixed length**: Read-only. If the driver expects chunks of data with a fixed length, they can define it here. This field displays the fixed length of the chunks in bytes and is optional. Details on individual instruments can be found under **Help > Instrument Setup**.

3.2 Edit I/O settings

To edit or view the instrument interfacing, select the instrument in the Properties view and then click the **…** button to the right side of I/O. The following dialogue box will open.

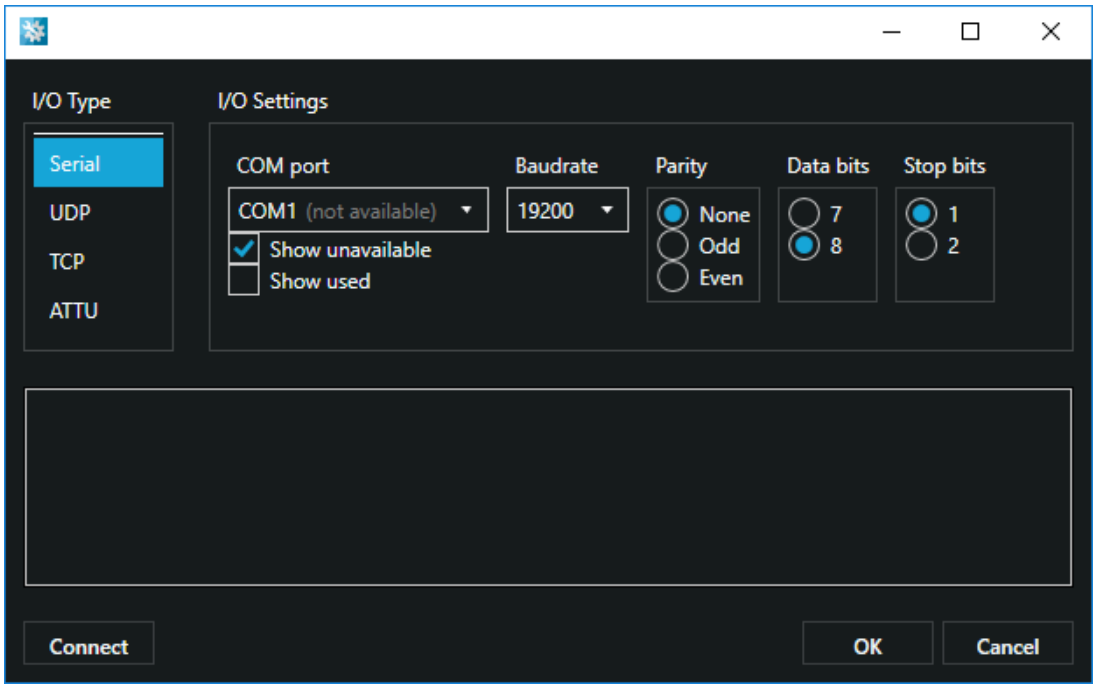


Figure 20 Instrument I/O dialogue box

Some properties are copied over from the properties pane in the main window, but some I/O-specific properties are only accessible here:

- **I/O Type**
  Select how to interface the instrument
  - **Serial**: Ordinary COM port.
  - **UDP/IP**: Network via UDP/IP datagram.
  - **ASCII File**: (For data output instruments only) Generate ASCII file on **\eiva\navipac\data**.
  - **ATTU**: Network interfacing via ATTU/TimeBox.
  - **TCP/IP**: Network via server connection (NaviPac acts as client).
- **I/O Setup**
  Properties related to specific I/O type
  - **Serial**: **Port**, **Baudrate**, **Parity**, **Stopbits** and **Databits**.
  - **UDP/IP** and **TCP/IP**: **Port** and **IP address**. For incoming data, the IP address must be on the sending computer or just 0.0.0.0.
  - **File**: ASCII file name
  - **ATTU**: **Port** and **IP address**.
- **Connect**
  For serial and network input only – open port and display raw data

3.3 Edit special settings

Some instrument types require a very detailed setup. These special setup properties can be accessed by right-clicking on the specific instrument in the **Tree** view. In the pop-up menu that opens, select **Edit special settings**, and a dialogue box will open. Special properties apply to the following instrument types, including (among others):

- User-defined output
- Digicourse
- DOF SubSea pipe inspection

A list of individual instrument types that allow for special properties editing is available in the NaviPac Instrument Parameters **Help** file, which can be accessed under **Help > Instrument Setup**.

3.4 Remove instrument

To remove an instrument, right-click on the specific instrument in the **Tree** view or the instrument box in the **Vehicles** view. In the pop-up menu that will open, select **Remove instrument**.

3.5 Add instrument

Adding an instrument can be performed in multiple ways

- Right-click on the **Vessel** top level item in the **Tree** view or **Vehicles** view and in the pop-up menu that will open, select **Add instrument**.

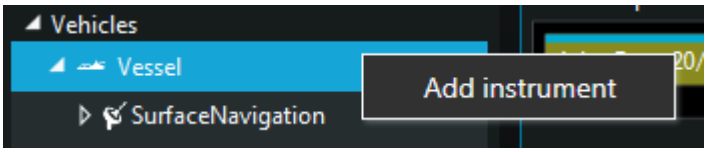


Figure 21 Adding new instrument from top level

- Right-click on the instrument group (eg **Motion sensor**) in the **Tree** view and in the pop-up menu that will open, select **Add new item**.

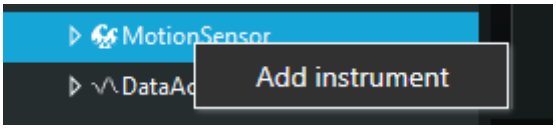


Figure 22 Adding new instrument from instrument group level

If you have started the process from the top level, selecting **Add new item** will open a selection dialogue box including all instruments:

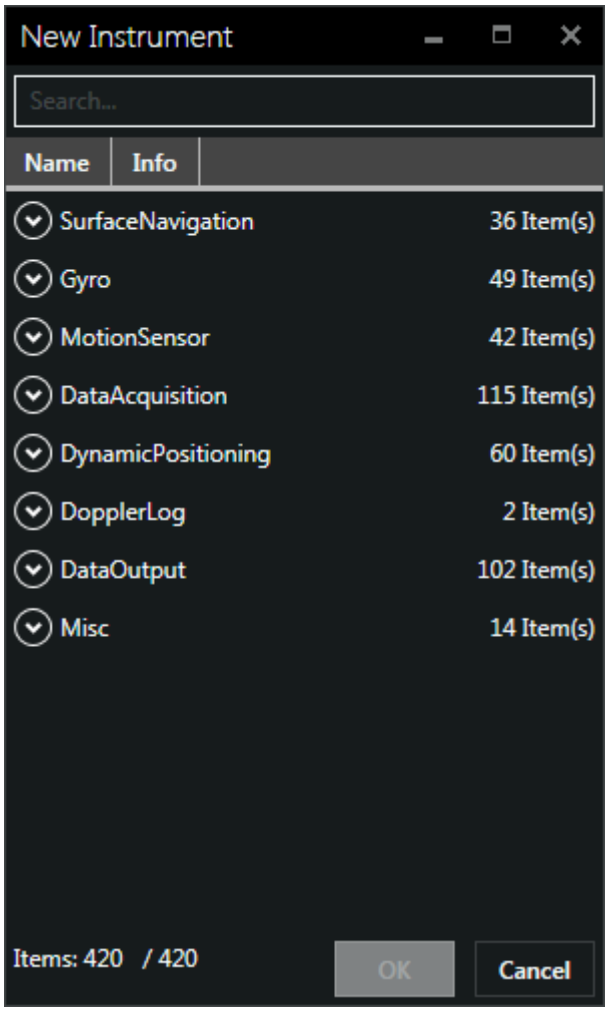


Figure 23 Selection of instrument group

Here, in the **New Instrument** dialogue box, you may expand the group you want to use, or search across groups.

Expanding a group gives same result as just selecting to add an instrument of the specific type:

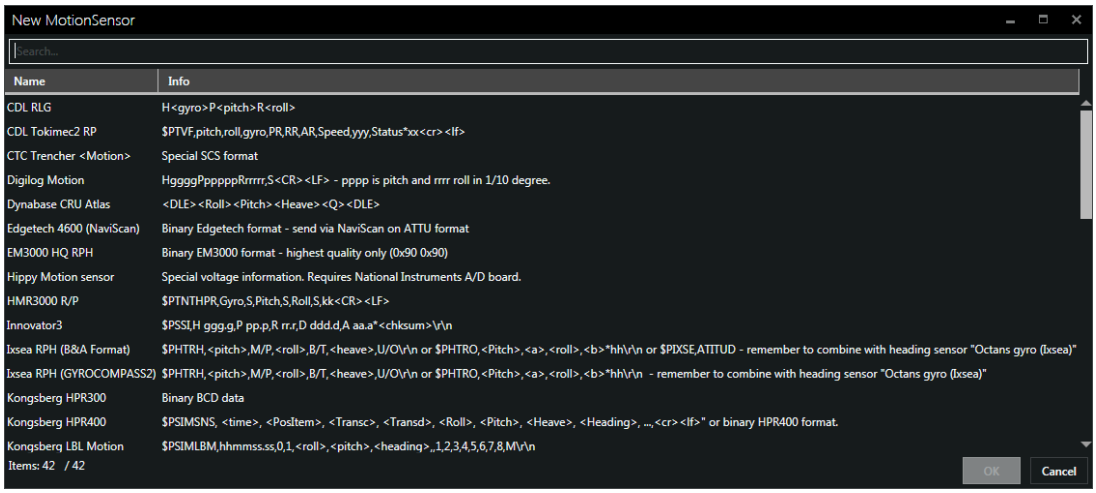


Figure 24 Selection of the specific instrument type

The list of instruments shows the names as defined by the operator, and the expected data format. You can search in the list by typing key words in the search field, as seen in the figure below.

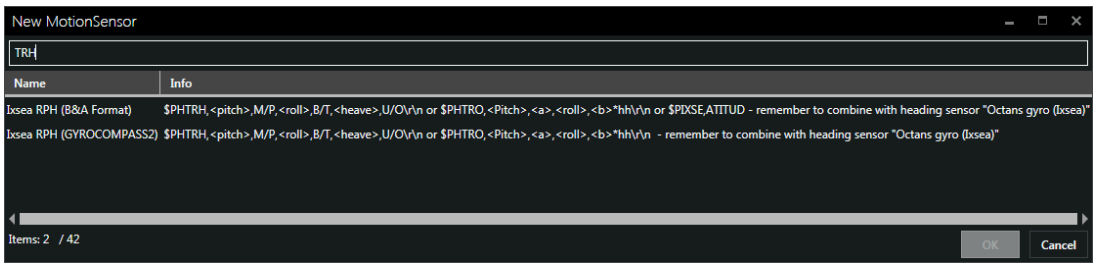


Figure 25 Use of the search field

You must then select an instrument and accept by clicking **OK**.

Next, the ordinary **Instrument I/O** dialogue box opens, as described in this manual in section 3.2 'Edit I/O settings'. Click **OK** when you are done editing to apply changes made.

Lastly, the new instrument will now appear in the **Tree** view, and can be selected in order to finalise the editing of its properties in the **Properties** view, as described in this manual in section 3.1 'Edit instrument'.

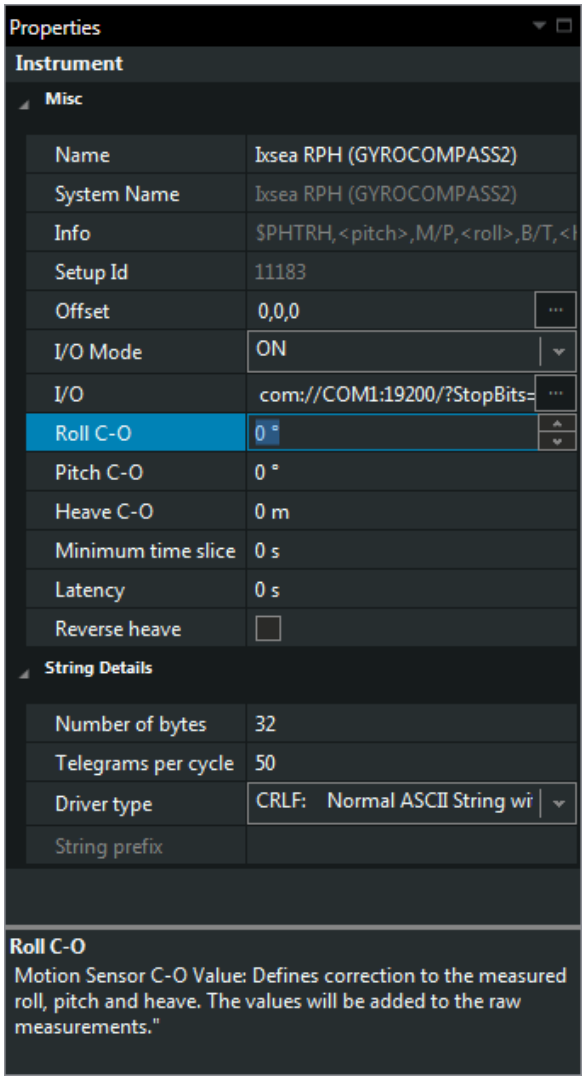


Figure 26 Finalising the instrument by editing its properties

3.6 Sort instrument list

NaviPac automatically sorts the instrument list within each instrument group in the **Tree** view based on the following criteria:

- Instruments set to **I/O Mode**: **Off** are kept at the bottom of each list.
- For instruments that are assigned to objects (gyro, motion sensor, Doppler log and data acquisition instruments only) sorting is thereafter done based on the object selected under **Location**. For example, if the ordering of objects in the tree view is Vessel, ROV1, Towfish, a gyro instrument set to **I/O Mode**: **On** (which under **Location** is assigned to **Vessel**) will be displayed higher in the gyro instrument group list than a gyro instrument set to **I/O Mode**: **On** (which under **Location** is assigned to **ROV1**).

If the **I/O Mode** or **Location** is changed for one or more instruments in an instrument group, the list order can be updated either by saving the current setup file or by right-clicking on the instrument group (eg **Motion sensor**) in the **Tree** view and, in the pop-up menu that opens, selecting **Sort list.** The current setup file must be saved in order for the list order to be saved and applied the next time you open NaviPac Configuration.

However, the user must specify which instrument is the primary instrument (eg the best gyro, best motion sensor, etc) in each instrument group. This can be performed by moving the instrument by the mouse in the vehicle view, as seen in the below figure.

Figure 27 Moving GPS2 in front of GPS1

3.7 Dynamic positioning

Some dynamic positioning systems (like USBL or LBL) support multiple tracking objects. These instruments will be defined in multi-step operations:

- Select the main vessel
- Click the **+** icon at the upper right corner in the **Vessel** window
- Expand **Dynamic Positioning**
- Select the sensor (eg HPR 410/HiPAP) you want to add
- Accept by clicking **OK**

- Select the correct interfacing setup
- The instrument is added to the list and selected (sensor details are shown at the right of the **Properties** window)
- If desired, enter a more reasonable name for the instruments to help identify the right sensor at a later stage
- Enter the offsets from CRP to transducer reference point (X across vessel + starboard, Y along vessel + front, Z + up)
- Enter USBL hold time (default: 5 seconds) to define data timeout
- Enter correct settings for USBL hold time (5 seconds or more) and USBL mountings
- **Note**: Please consider carefully how time tagging of the sensor is performed (eg refer to the NaviPac manual on timing principles)
- Point to **0** under Dynamic Positioning, which will turn into a **+**
- Move the mouse to the box and point to the **0**, which becomes a **+**

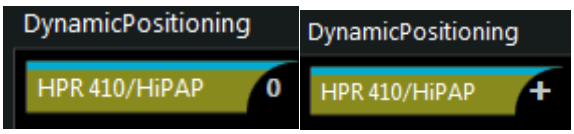


Figure 28 Positioning a ROV via USBL

  - Click on **+**
  - Select the name of the ROV in the list
- A USBL entry is now added to the ROV instrument list and linked to the vessel USBL
- Select the ROV USBL item
- Enter a transponder code corresponding to the settings in the USBL system
- Enter other USBL-specific parameters if needed

4 Geodesy

The currently selected geodesy can be viewed and controlled from the **Tree** view **Geodesy** item:

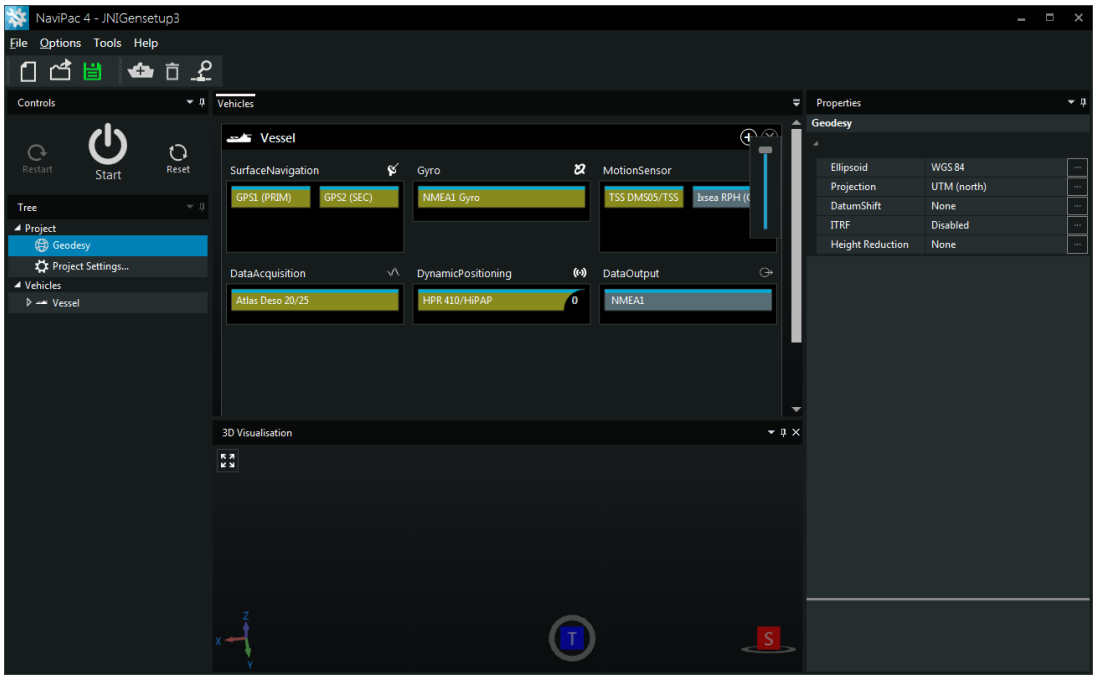


Figure 29 Display of current Geodesy setup

The geodesy consists of 5 items which can be configured individually via the **…** button at the right and which includes

- **Ellipsoid**
- **Projection**
- **Datum Shift**
- **ITRF**
- **Height Reduction**

4.1 Projection and ellipsoid

Ellipsoid and projection is both edited in the same environment:

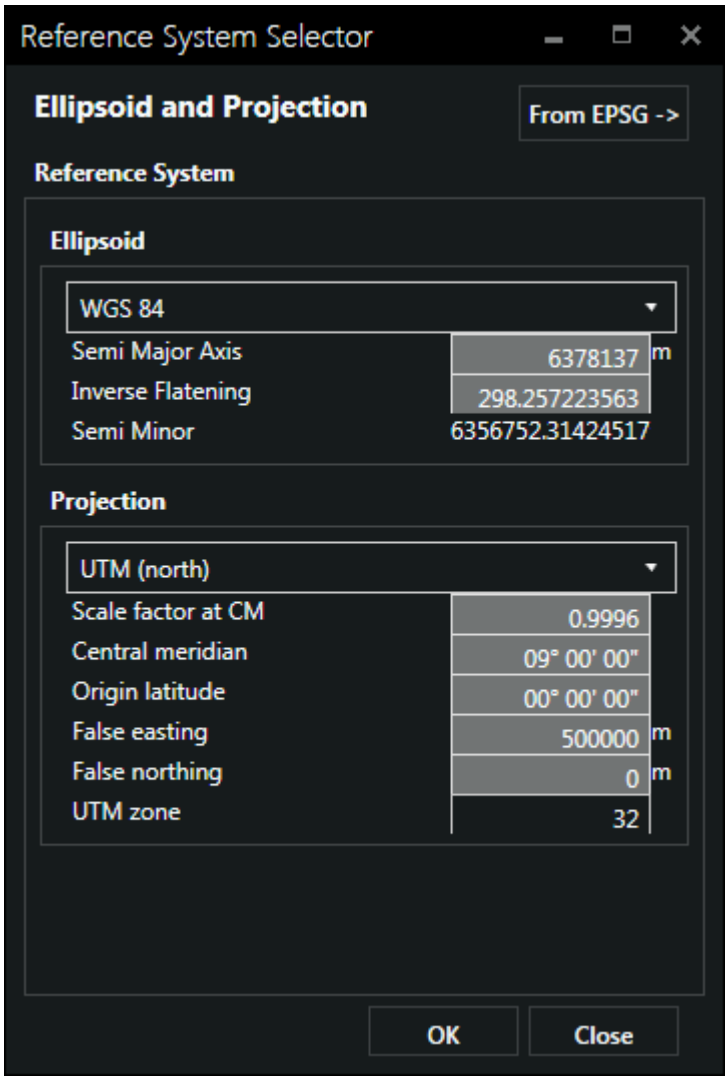


Figure 30 Selection of ellipsoid and projection type

This **Ellipsoid and Projection** dialogue box is able to read the geodetic settings from the setup file and then display it in fields here. The available choices of predefined **Ellipsoid** and **Projection** are read from the setup file and inserted into two drop-down lists, as shown above.

When the user selects an ellipsoid/projection, the relevant information for the selection is displayed in fields below the selection. Some fields will be read-only, and some projections

will have more fields than others, etc. Selections and changed values will be applied upon clicking the **OK** button.

In the above example, the projection **UTM (north)** and the **WGS84** ellipsoid have been selected. Changes made in this dialogue box will be saved in the setup file.

Selecting a different projection than the current one will change the rest of the fields in the dialogue box, as the properties depend on the selected projection.

Ellipsoid: Drop-down list with all available ellipsoids. The values in the three fields listed below will be set automatically on the basis of this selection:

- **Semi Major Axis**: Displays the semi major axis of the ellipsoid in metres (eg ED50 is 6378388.0000 metres).
- **Inverse Flattening**: Displays the inverse flattening of the ellipsoid. Flattening is calculated as:
  $\frac{1}{InverseFlattening}$ (eg ED50 is 297.0000000000 metres)
- **Eccentricity**: Displays the eccentricity of the ellipsoid. Eccentricity is another way of defining the flattening of the ellipsoid, as eccentricity equals:
  $\sqrt{Flattening*(2-Flattening)}$ (eg ED50 is 0.081991889979 metres)

Projection: Drop-down list with all available projections. The selection of available properties listed below will depend on the specific projection in use (similar properties with different names may be available instead, depending on the system). Some fields are read-only and some are editable:

- **Scale factor at CM**: The point scale factor at origin of projection (central meridian)
- **Central meridian**: The longitude at origin of projection
- **Origin latitude**: The latitude at origin of projection
- **False easting**: The **False easting** applied to calculations
- **False northing**: The **False northing** applied to calculations
- **UTM zone**: The **UTM zone** number

4.1.1 EPSG

NaviPac includes the most recent EPSG database, and you can use that as a lookup table via the **From EPSG** button:

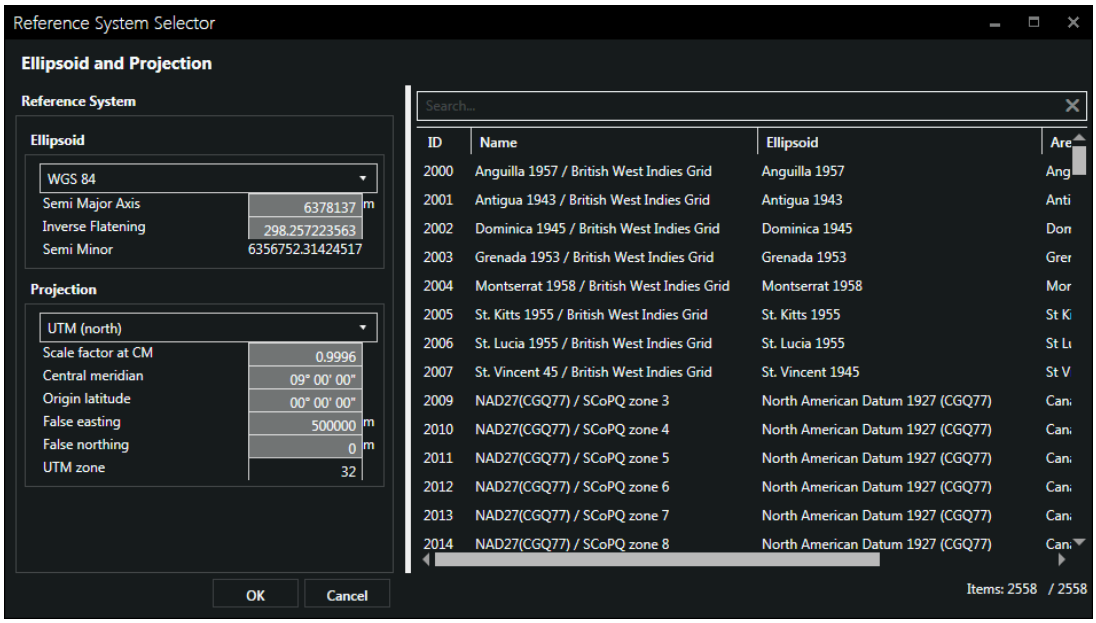


Figure 31 Searching in EPSG database

The EPSG base is currently only supporting transverse-Mercator-based methods.

4.2 Datum shift

Datum shift will act differently depending on whether or not you are using EPSG as the basis for the projection and ellipsoid or not. When using EPSG for the projection selections, you may also use it in the datum shift selection

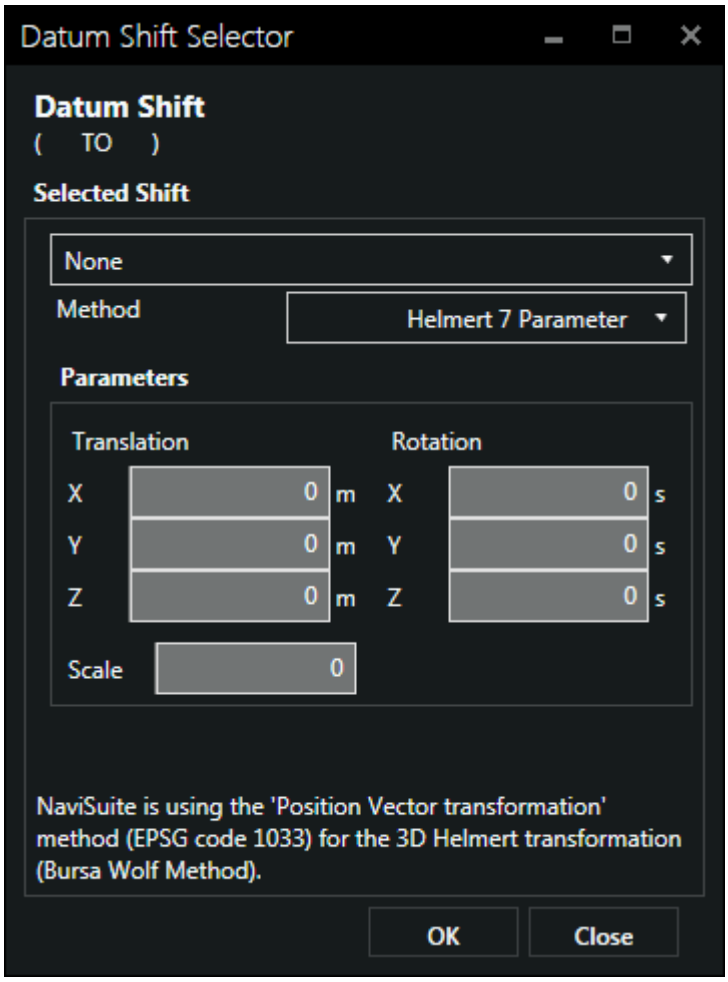


Figure 32 Defining datum shift without EPSG

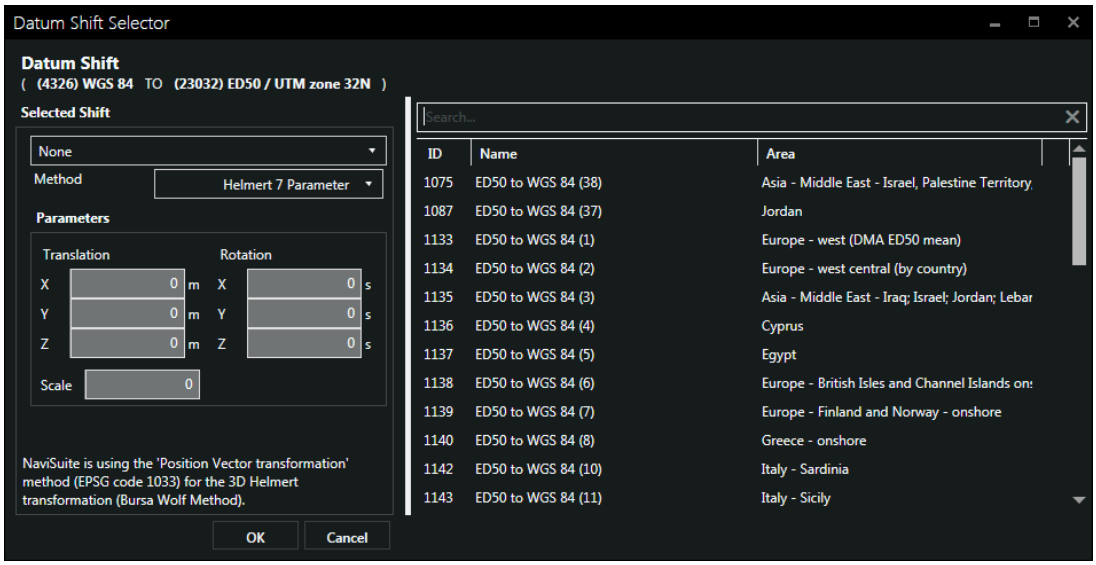


Figure 33 Defining datum shift when using EPSG

The dialogue box that opens allows for specification of datum shift data (7-parameter transformation from WGS84 to user datum or US NADCON).

Note: Datum shift must be enabled manually for each individual GPS instrument under the **Apply datum shift** property in the general instrument setup.

The seven parameters are read from the setup file and displayed in the fields in this dialogue box. For all predefined shifts, the seven parameters will display predefined values; however, if the user selects **User Defined Datum Shift** from the **Name** drop-down list, all seven parameters must be manually defined, using up to 12 significant digits per parameter:

- **TX** (default in metres)
- **TY** (default in metres)
- **TZ** (default in metres)
- **RX** (default in seconds
- **RY** (default in seconds)
- **RZ** (default in seconds)
- **Scale** (parts per million)

The parameters will be used in one of the following calculations:

4.2.1 Helmert 7

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} Tx \\ Ty \\ Tz \end{pmatrix} + \begin{pmatrix} PPM & -Rz & Ry \\ Rz & PPM & -Rx \\ -Ry & Rx & PPM \end{pmatrix} \times \begin{pmatrix} X0 \\ Y0 \\ Z0 \end{pmatrix} + \begin{pmatrix} X0 \\ Y0 \\ Z0 \end{pmatrix}$$

4.2.2 Mod. Bursa-Wolf

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} Tx \\ Ty \\ Tz \end{pmatrix} + \left(1/(1-PPM)\right) * \begin{pmatrix} 1 & -Rz & Ry \\ Rz & 1 & -Rx \\ -Ry & Rx & 1 \end{pmatrix} \times \begin{pmatrix} X0 \\ Y0 \\ Z0 \end{pmatrix}$$

4.2.3 NADCON US

Between NAD83 and NAD27, where NAD83 is to be seen as WGS84. The NaviPac implementation of NADCON (North American Datum Conversion Utility) is based on ASCII files for the American zones, and is an empirical solution rather than a mathematical solution. The NADCON files define a grid in latitude/longitude with transformation constants for each grid point. NaviPac uses this as a lookup table and interpolates between the four closest points to get actual shift correction. See http://www.ngs.noaa.gov/PC_PROD/pc_prod.shtml#NADCON for further details and http://www.cs.arizona.edu/icon/ftp/data/nadcon for a list of available grid files.

4.3 ITRF

With ITRF, you may define whether NaviPac should include time-based datum shift parameters to account for situations where the datum shift is changing over time. If you choose to enable ITRF, the basic NaviPac datum shift parameters will be adjusted daily using the **Speed of change** parameters, based on the reference date.

This type of datum shift can, for example, be relevant in North America, where the use of ITRF to NAD83 is widely used, and in Europe to shift from ITRF to EUREF89.

As the ITRF parameters are adjusted 'on-the-fly', please use with care.

Please note that there is a dedicated datum shift toggle for each GPS instrument in use (the **Apply datum shift** property in the instrument's general setup), and you should disable this if you have already compensated for this shift in the GPS instrument.

ITRF parameters are entirely defined in the properties pane after selecting **ITRF** from the **Tree** view:

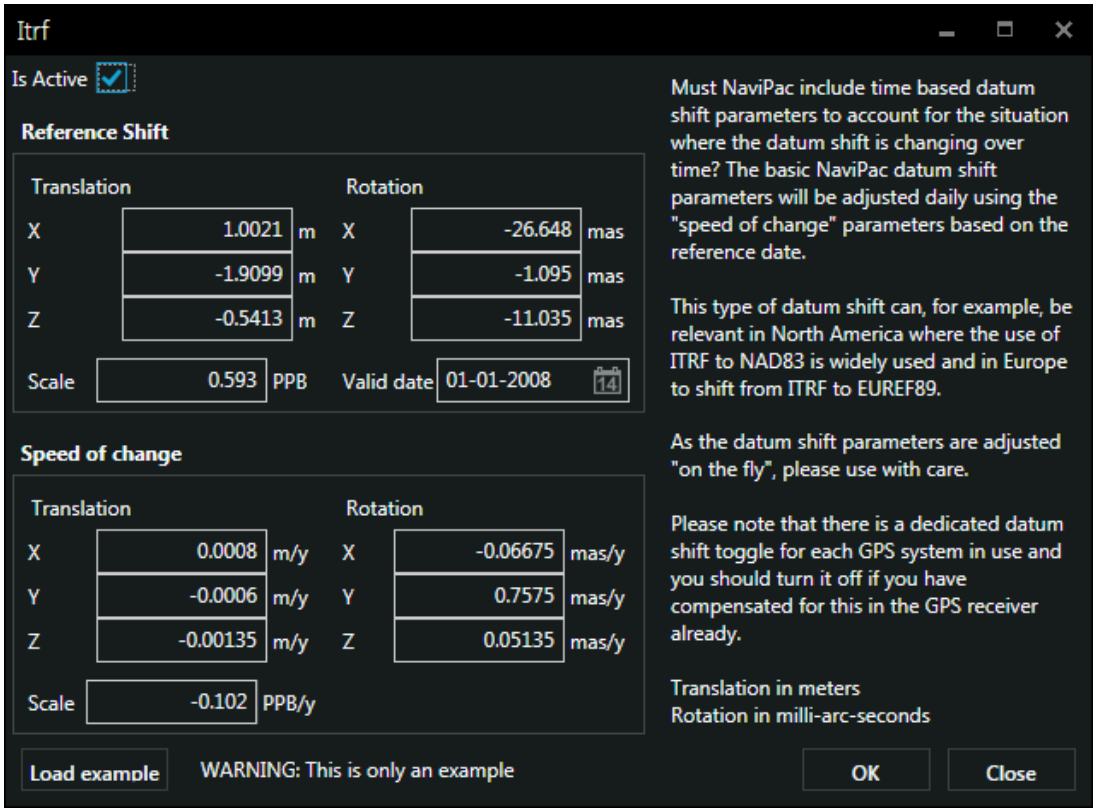


Figure 34 Definition of ITRF Reference Shift and Speed of change parameters

ITRF consists of the following parameters:

- **Reference Shift**: Specify a traditional 7-parameter shift together with a valid date. Figure 34 above defines the ITRF2008 to NAD83 that was valid on 1 January 2014. (New reference sets may be defined.)
- **Speed of change**: Specify the changes of the defined seven parameters per year.

Transformations are measured in metres, rotations in milliarcseconds and scale in parts per billion.

If you select the **Is Active** check box at the top of the view, then you need to go in and enable this on the individual GPS receivers when required.

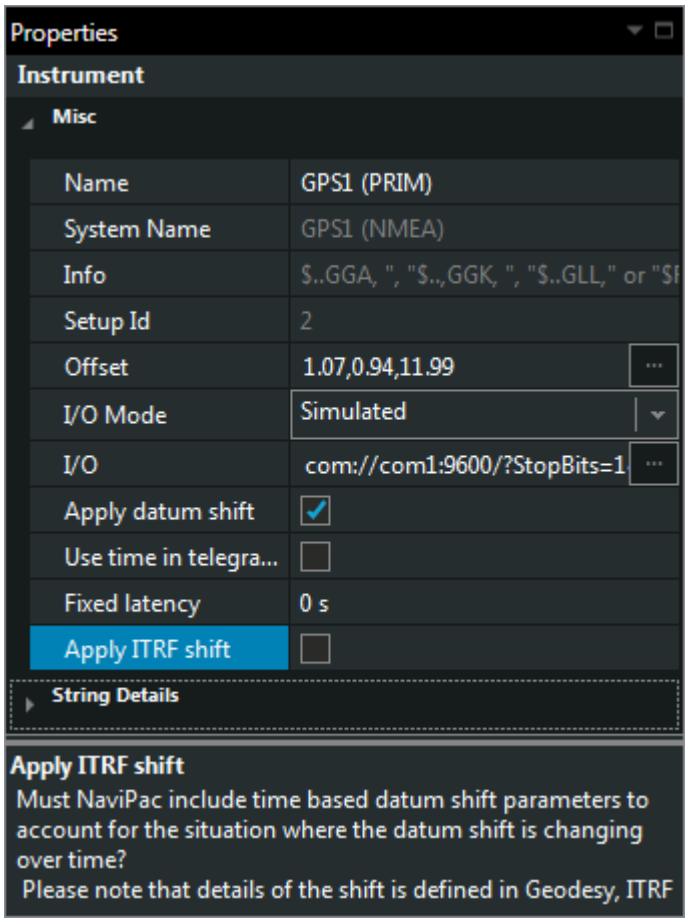


Figure 35 Apply ITRF Shift for an individual GPS instrument

Note: **Please use ITRF with care, and make sure that you test against a reference point.**

4.4 Height reduction

Height reduction defines a correction for the used GPS height similar to what datum shift does for the 2D position. In the short term, this determines how the operator gets from WGS ellipsoid to the local reference system.

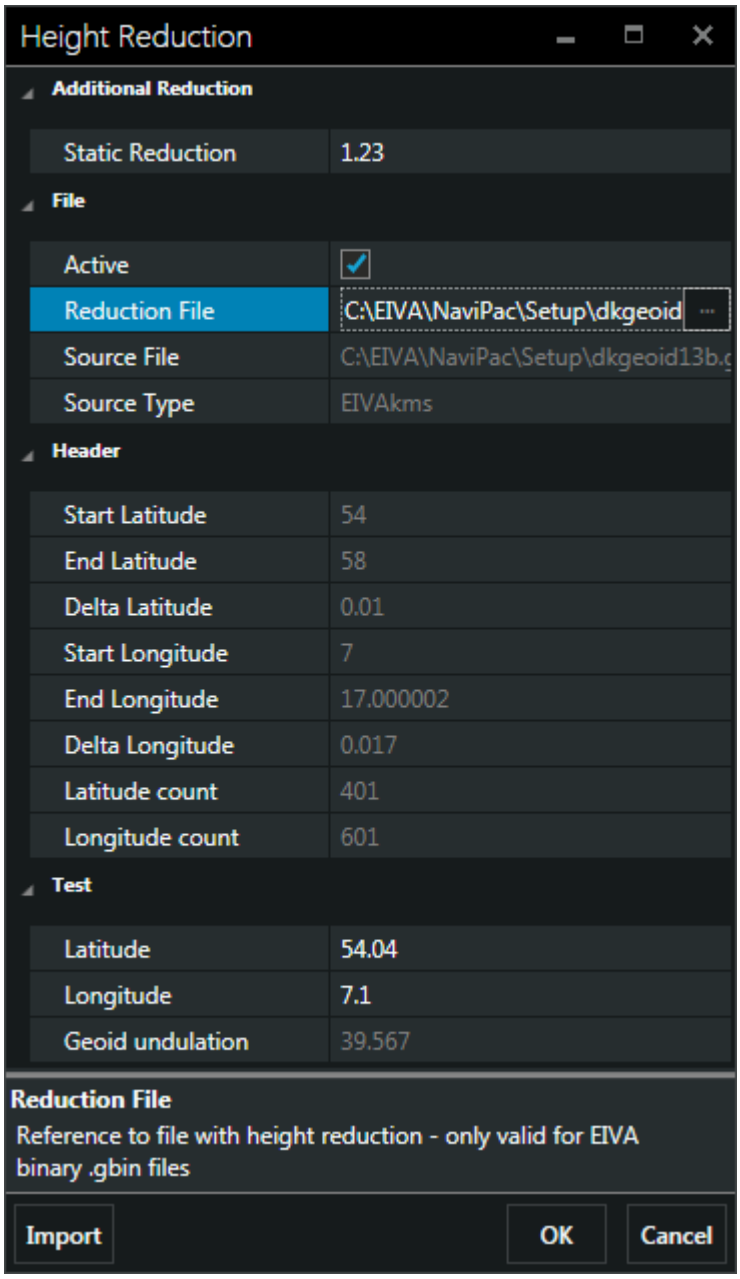


Figure 36 Height reduction

Height reduction can either be a manual offset or a method-based solution (ie VORF or USGEOID09) or a combination.

Details can be found in the dedicated geodetic reference manual, which can be found in the **Help** menu.

Note: Please use with care, and make sure that you test against a reference point.

5 Project settings

The project settings are a collection of global parameters in the project, and they can be accessed and addressed by clicking **Project** in the **Tree** view

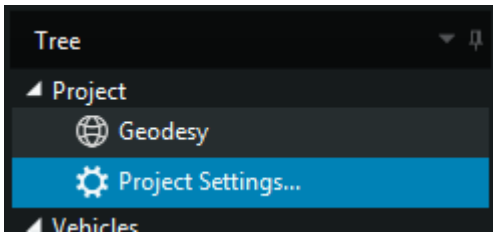


Figure 37 Activating project setting

The setting consists of a series of panels as discussed in this chapter.

5.1 NaviPac

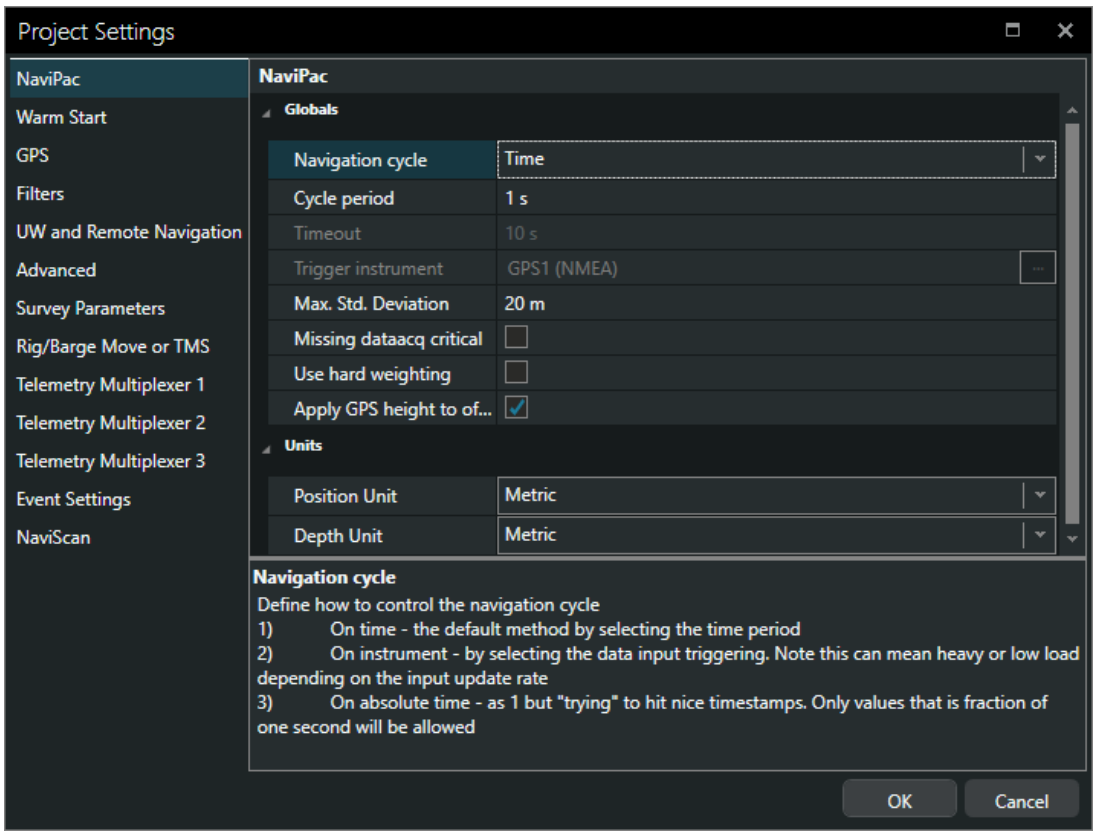


Figure 38 Parameters for NaviPac kernel

Navigation cycle
Define which cycle mode NaviPac should operate in

- **Time**: Asynchronous cycle based on time period
- **Input**: Trigger by input
- **Absolute time**: Semi-asynchronous cycle based on time period

Timeout
(For **Input** cycle mode only) Define the maximum allowable timeout length in cases of missing input.

Trigger instrument
(For **Input** cycle mode only) Select the instrument that will trigger the cycle.

Cycle period
Define how often NaviPac should calculate a reference position. The default value is **1 s**.

Max. Std. Deviation
During runtime, NaviPac calculates a standard deviation of the position (in case of redundancy in position). This calculation gives a value for how correct the position is. To monitor the correctness, NaviPac generates an alarm when the standard deviation exceeds the defined maximum limit. Reasonable limits could be:

- Highest/Fixed RTK < 0.5 m
- Differential < 2 m
- All (Autonomous) < 15 m

Missing dataacq critical
Select whether you want NaviPac to go into red state (by selecting this check box) if no data is received from a data acquisition instrument (eg an echo sounder), or whether you just want alarms to be generated (by clearing this check box / leaving this check box blank).

Use hard weighting
Select whether you want NaviPac to perform a soft (ordinary) or hard (rejects sample completely if more than 2 sigma away) weighting of navigation data when operated with a single GPS. Hard weighting will provide better error rejection, but will also increase the risk of NaviPac going into red state.

Apply GPS height to offsets
If the GPS is in **Highest/Fixed RTK** mode, select (by selecting this check box) whether all offsets must be calculated relative to the resulting GPS height (orthometric height/local datum) or relative to local vessel 0 (by leaving this check box blank / clearing this check box).

Units
NaviPac operates as default in metric mode – but here you can select to us eg US survey feet or international feet

5.2 Warm start

One important setting (especially in simpler configurations) is the selection of modules the user wants to see during online navigation startup. When NaviPac starts the navigation cycle, the user can specify which modules should open automatically. This can be controlled from **Warm start**:

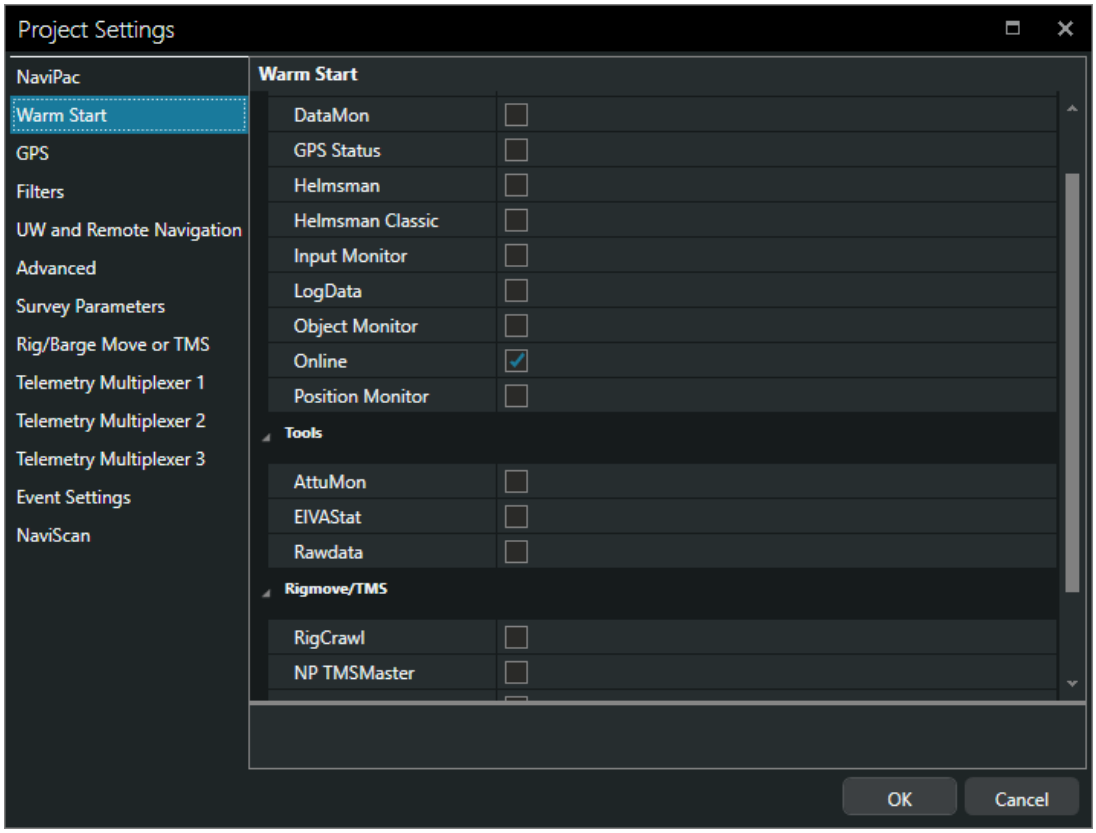


Figure 39 Selection of modules to include in Warm start

You may add your own modules by editing the **Custom program** field (each program must be written by a full path).

5.3 GPS

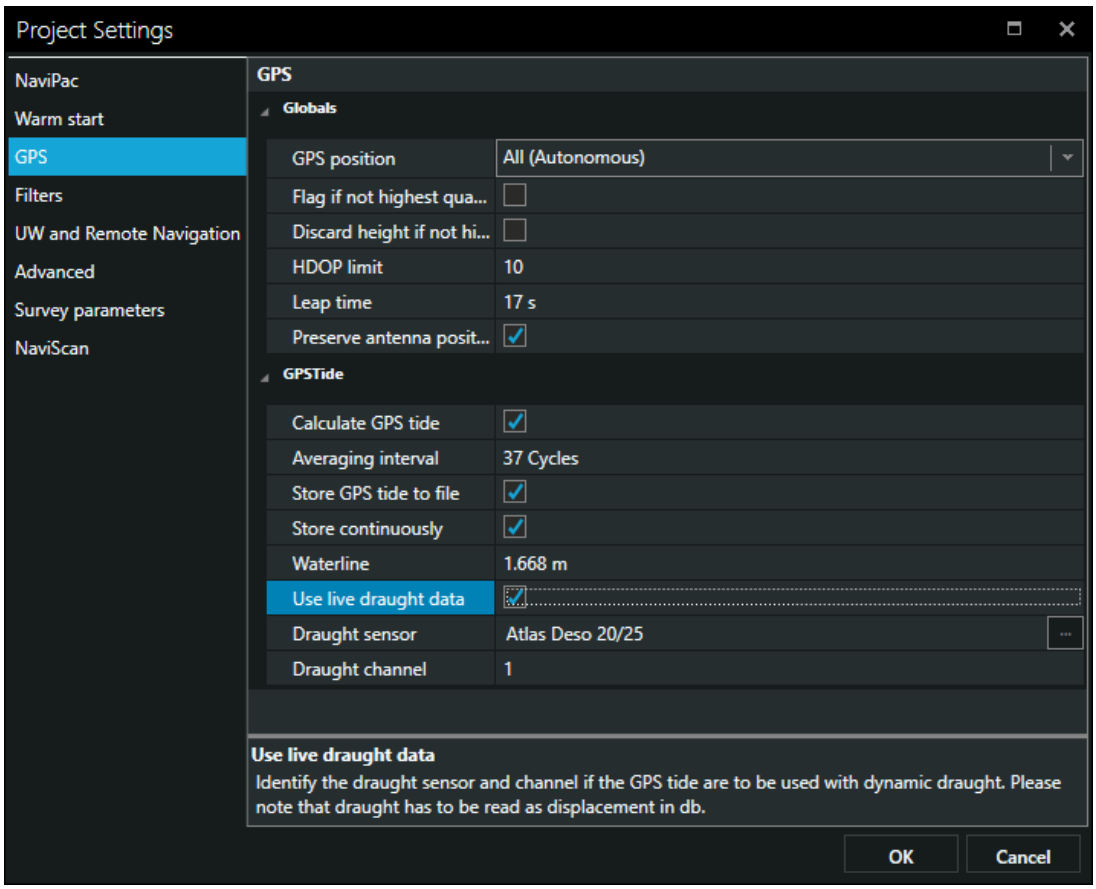


Figure 40 Global GPS definitions

GPS position
The GPS can output position in various quality levels (fix status, etc). In NaviPac, this is mapped on three levels:

- **All (Autonomous)**
- **Differential**
- **Highest/Fixed RTK**

For this parameter, you can select the level(s) of GPS quality that you will accept.

Note: Please check that your GPS data is interpreted correctly before using this filter.

Flag if not highest quality
If you want to use GPS even without the highest quality, but need to make it clear that the

quality is lower, you can choose to let NaviPac flag the data as yellow state (use this function with caution) by selecting this check box.

Note: Please check that your GPS data is interpreted correctly before using this filter.

Discard height if not highest quality
If you only want to use GPS height with the highest quality, you can select that only **Highest/Fixed RTK** data will be used in height (Z) calculations by selecting this check box.

Note: Please check that your GPS data is interpreted correctly before using this filter.

HDOP limit
For this parameter, you can define a maximum allowed GPS horizontal dilution of precision (HDOP). If the value becomes higher than the defined limit, an alarm is generated, and the GPS position is weighted down.

Leap time
This parameter allows the user to specify the difference between the GPS clock and UTC time (ie time to be subtracted from the GPS time to get the UTC time). As of 30 June 2015, the leap time was 17 seconds.

GPS time (Global Positioning System time) is the atomic time scale implemented by the atomic clocks in the GPS ground control stations and the GPS satellites themselves. GPS time was zero at 12 am on 6 January 1980, and since it is not perturbed by leap seconds, GPS time is, as of 30 June 2015, ahead of UTC time by 17 seconds.

Preserve antenna positions
Use this check box to select whether you want NaviPac to treat the raw GPS antenna position (without any corrections, skewing, etc) as a separate object (ie making it visible on the display, in logging, etc).

Calculate GPS tide
Select whether to use GPS height to calculate GPS tide. The primary GPS height will be compensated for antenna offset, roll, pitch and heave (thus representing normal height of the vessel CRP).

Averaging interval
To get rid of noise, the calculation uses an averaging interval (eg 30 cycles). NaviPac calculates a GPS tide value at each cycle and puts it into a list. The corresponding values will hereafter be the average of this list. The timestamp will be the average time of the data in the list.

Store GPS tide to file
NaviPac creates a tide file in the same folder as the ordinary recording. The files will be named ddmmyy.NPT, and NaviPac maintains one file per day. The data format will be:

$NPTide,<Date>,<Time>,<Tide>,<GPS H><cr><lf>

Tide is **positive** above datum.

Store continuously
Define whether NaviPac should store one measurement per cycle or reduce to the selected averaging interval. A typically scenario would use a 30-second averaging interval, so it is a matter of whether you want to have one value saved in the file per second or one value saved in the file per 30 seconds.

Waterline
Offset between vessel CRP and waterline.

Use live draught data
Draught sensor
Draught channel
Identify the draught sensor and channel if the GPS tide are to be used with dynamic draught. Please note that draught must be read as displacement in db.

5.4 NaviPac filters

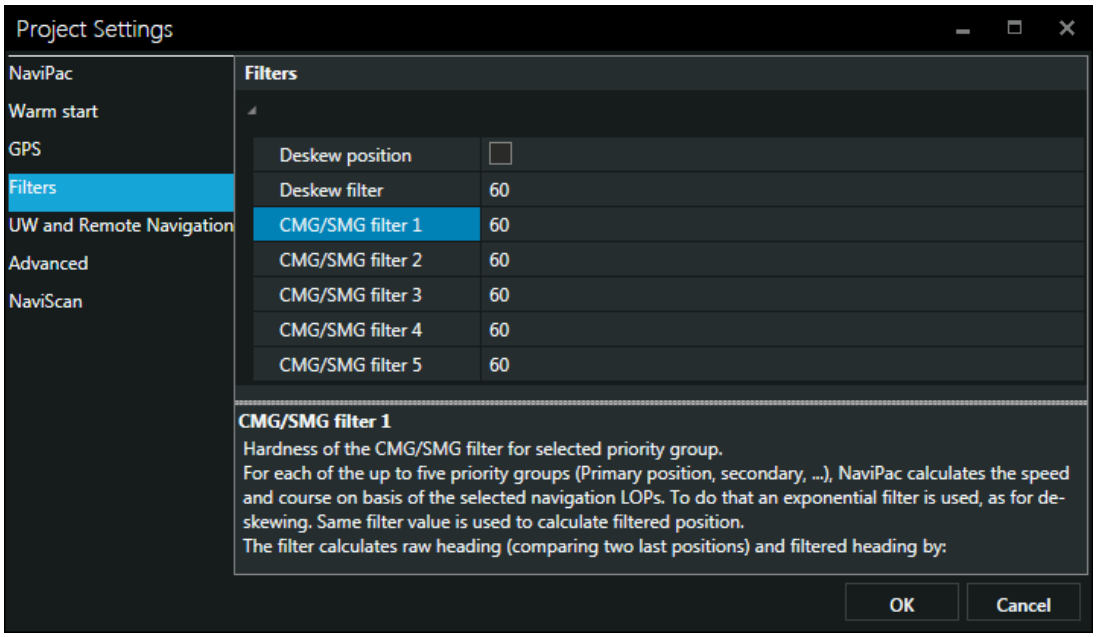


Figure 41 Defining general filters

Deskew position
Select whether you want surface navigation data from the position sources to be deskewed to the correct time or not. This check box must be selected in order to be able to combine and compare multiple surface navigation systems. See this guide dedicated to deskewing

questions for further details:
https://eiva.freshdesk.com/support/solutions/articles/43000568159-deskew-position-data.

Note: For most survey operations we recommend that **Deskew position** is not checked.

Deskew filter
NaviPac uses a skewing technique to adjust measurements to the same time – in other wordsif you have two independent GPS systems, you must correct for age and delays in order to compare the two. To do that, NaviPac uses an exponential filter to calculate filtered value **ValF**.

First, NaviPac calculates a smooth **rate of change** (or speed) of the sensor by an exponential filter:

*ValF(T) = (1-G)*Val(T) + G*ValF(T-1)*

Filter value:

G = exp(-CycleFrq/Filter)

Then, NaviPac calculates the predicted value:

*VT= VTgps+ Vfilt*Age*

Using ValF, NaviPac can calculate smooth speed and acceleration of each position source in use and thereby correct for any delays and age. The value should be set based on the size of the vessel (ie how fast it can change course or speed. See the NaviPac Online manual for further details).

CMG/SMG filter 1–5
For each of the (up to five) priority groups (primary position, secondary position, etc), NaviPac calculates the speed and course based on the selected navigation position sources. To do this, an exponential filter is used, as for deskewing. The same filter value is used to calculate filtered position. The filter calculates raw heading (comparing the two last positions) and filtered heading by:

*F_H(T) = (1-G)*H(T) + G*F_H(T-1)*

Where *G* = exp(Cycle Frequency/Filter), *F_H* = filtered heading (ie CMG), and *H* = raw calculated heading.

The same principle is used for SMG.

5.5 Underwater and remote positioning

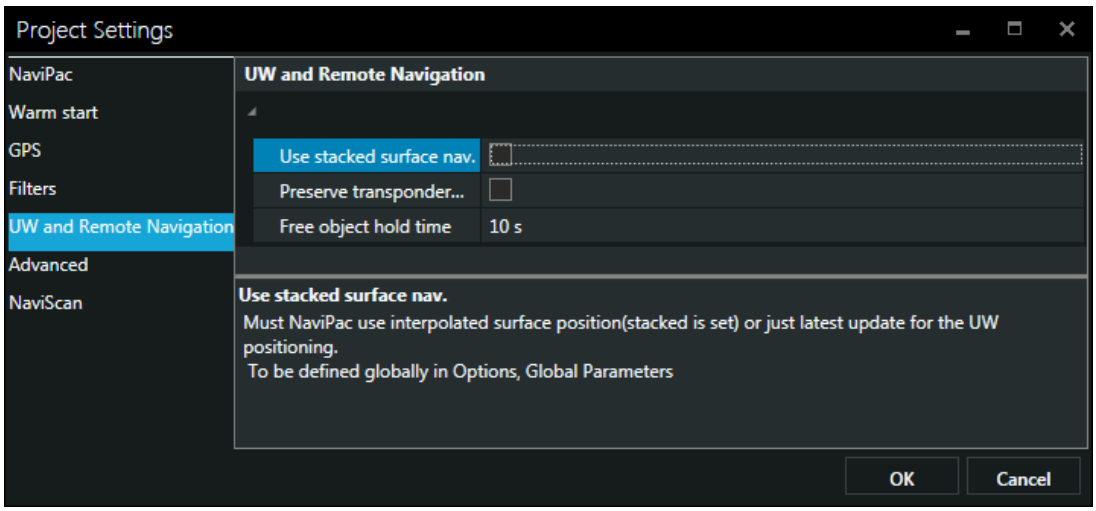


Figure 42 Global setting of remote positioning

Use stacked surface nav.
Select whether or not NaviPac should use interpolated surface position (stacked as set) or just the latest update for reference of the calculated underwater positioning.

Preserve transponder positions
Select whether or not NaviPac should create a second object for a new transponder position. This is only relevant for transponders where offsets are defined in the instrument setup. It will be ignored for those that have not been changed from the default value of **0 m**. This parameter is similar to the **Preserve antenna positions** parameter.

Free object hold time
Define how long NaviPac should keep free (unassigned) underwater objects active in the navigation measurements.

IMCA Id
IMCA ID (two characters giving company IMCA ID, eg Stolt Offshore = ST).

5.6 Advanced

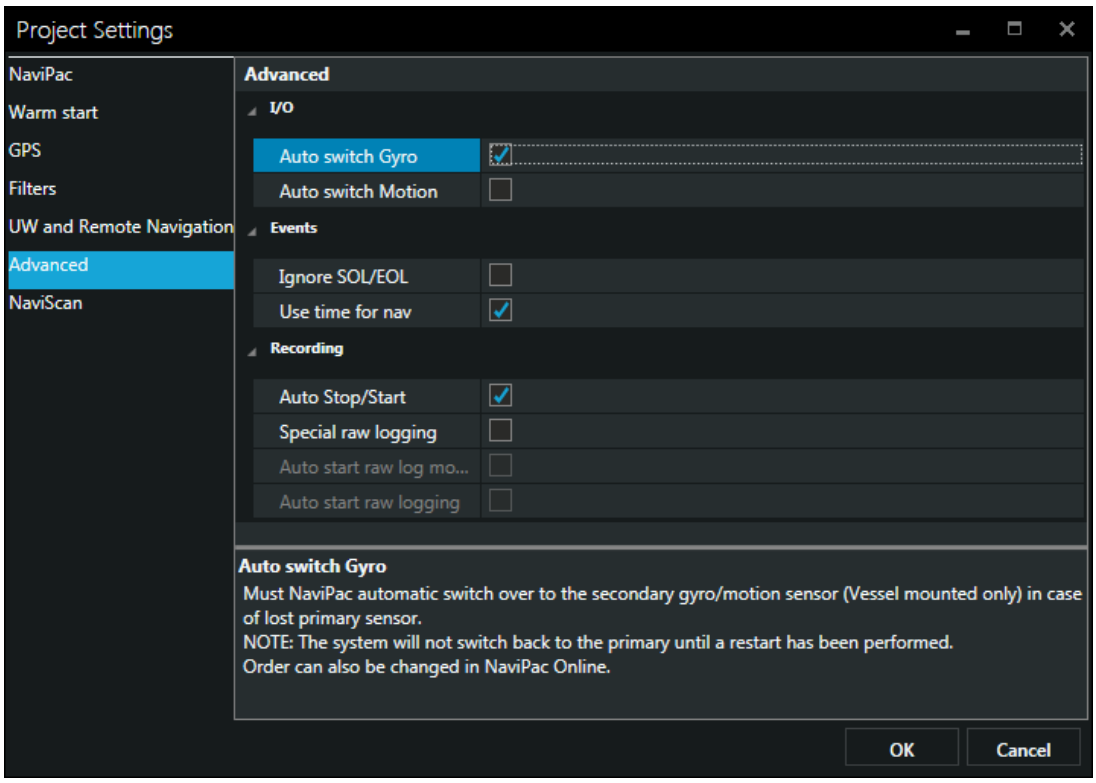


Figure 43 Collection of miscellaneous settings

Under this tab, you can edit a series of advanced parameters. Some of these may have significant influence on the behaviour of the software, so please handle with care. The parameters have been divided into categories in NaviPac to provide a better overview:

- **I/O**: **Auto switch Gyro**, **Auto switch Motion**
- **Events**: **Ignore SOL/EOL**, **Use time for nav**
- **Recording**: **Auto Stop/Start**, **Special raw logging**

Auto switch Gyro
Select whether you want NaviPac to automatically switch over to the secondary vessel gyro if connection or communications to the primary vessel gyro have been lost (by selecting the check box).

Note: NaviPac will not switch back to the primary gyro until you have restarted the navigation.

Auto switch Motion
Select whether you want NaviPac to automatically switch over to the secondary vessel motion sensor if connection or communications to the primary vessel motion sensor have been lost (by selecting the check box).

Note: NaviPac will not switch back to the primary vessel motion sensor until you have restarted the navigation.

Ignore SOL/EOL
NaviPac generates manual events (special case) at start and stop points of recording (or from the start line/stop line in the Helmsman's Display). This can be ignored in certain scenarios such as distance shooting. Select this check box to ignore these events.

Use time for nav
NaviPac will normally perform navigation cycles at a fixed interval (defined under **Cycle period**). In certain operations, it is preferable to force a navigation update at the time of an event (eg in distance shooting). Select this check box to force this option.

Auto Stop/Start
Select whether or not you want NaviPac to automatically switch to a new log file if changes have been made in the online setup (shift priority, weighting, corrections, etc). It is recommended to enable this option (by selecting this check box), as it ensures that log file headers and data are consistent.

Special raw logging
Select whether you want NaviPac to enable special raw data logging for playback. To learn more, see https://eiva.freshdesk.com/support/solutions/articles/43000567825-navipac-raw-recording-and-replay.

Note: this function requires substantial disc space and must be started from the **Online > Options > NaviPac Raw Log** menu item or included in the warm start.

Auto start raw log module
If you have enabled special raw data logging for playback under **Special raw logging**, select whether or not NaviPac should start the **NaviPac Raw Log** module as part of navigation start.

Auto start raw logging
If you have enabled special raw data logging for playback under **Special raw logging**, select whether or not NaviPac should start the raw logging immediately and perform 24/7 black box recording.

5.7 Survey parameters

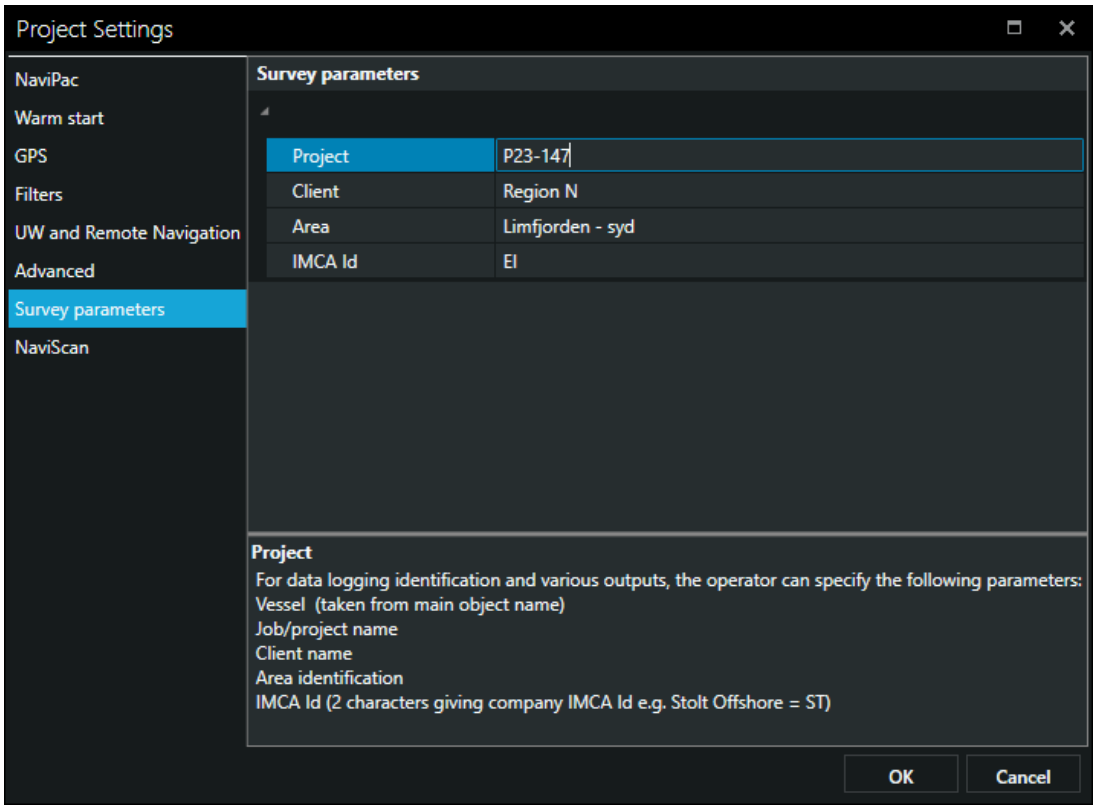


Figure 44 Defining survey parameters

For data logging identification and various outputs, the operator can specify the following parameters:

- Vessel (taken from main vessel object)
- Job/project name
- Client name

- Area identification
- IMCA Id (2 characters giving company IMCA Id e.g. Stolt Offshore = ST)

All parameters can be represented as 32 characters ASCII information, and will be presented in any general logging format files.
A few special outputs include the information.

5.8 Rig move

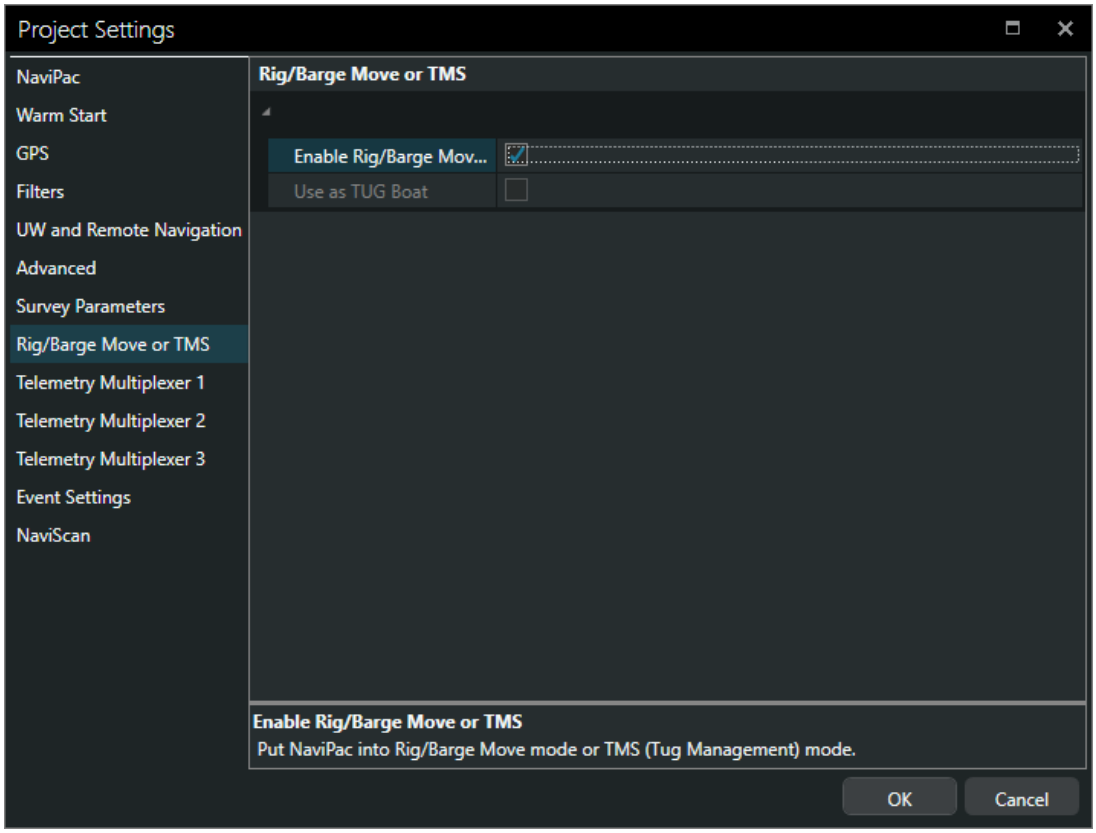


Figure 45 Special setting for Rig move

In this section you can enable NaviPac to use the Rig move and tug management setup. Please note that this section requires special NaviPac license

You can also mark that this NaviPac is used for tug boat handling

5.9 Telemetry setting

NaviPac can operate up to three telemetry radio modems to exchange information between vessels. The special multiplexer enable you to share incoming and outgoing data – and merge it into one serial data stream

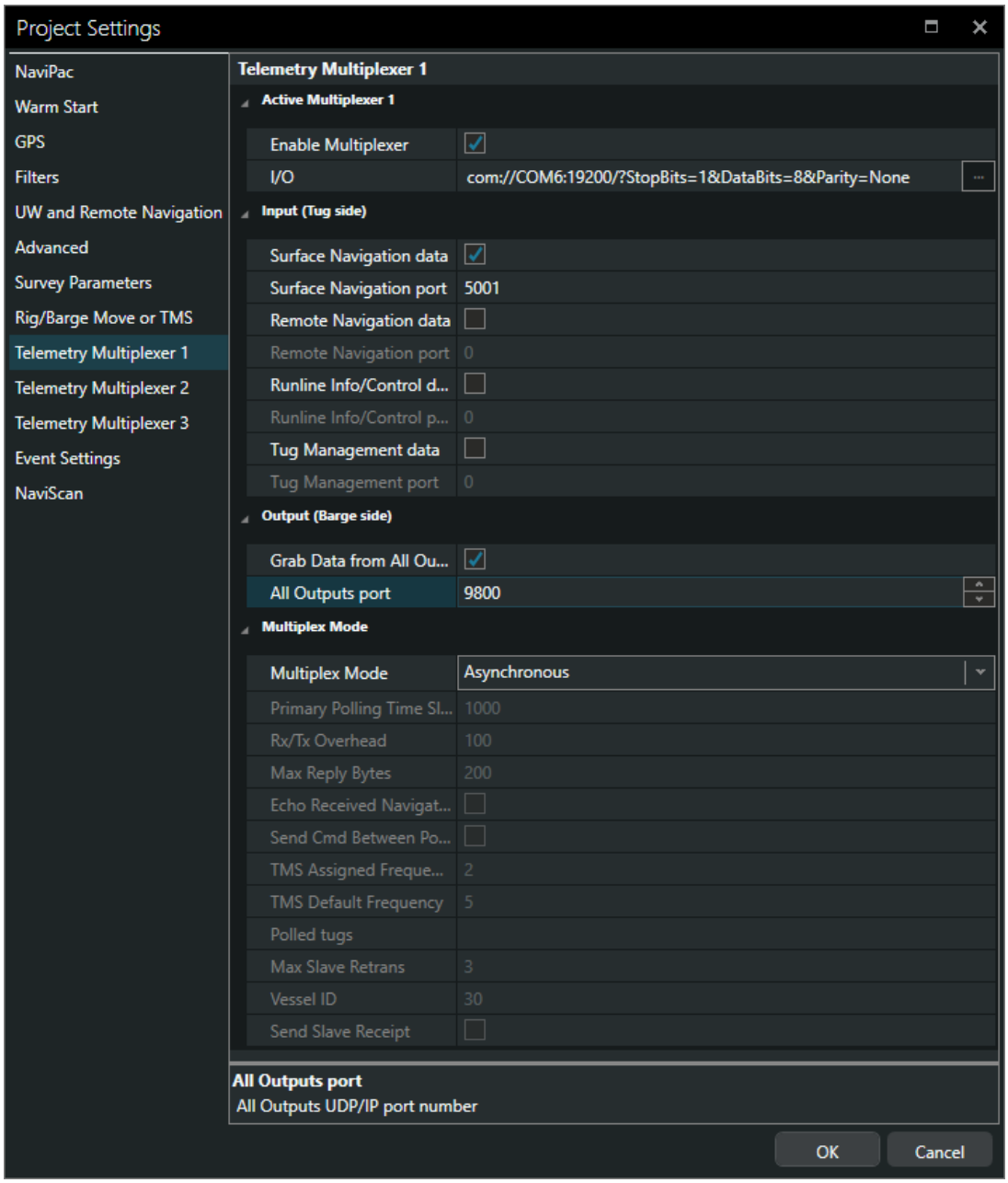


Figure 46 Multiplexer configuration

You must first of all identify the serial port (COM port) and communication parameters – please ensure they are set correct according to the radio modem

The second part identified how data is mapped from NaviPac (using UDP/IP packages on 127.0.0.1) for both incoming and outgoing data

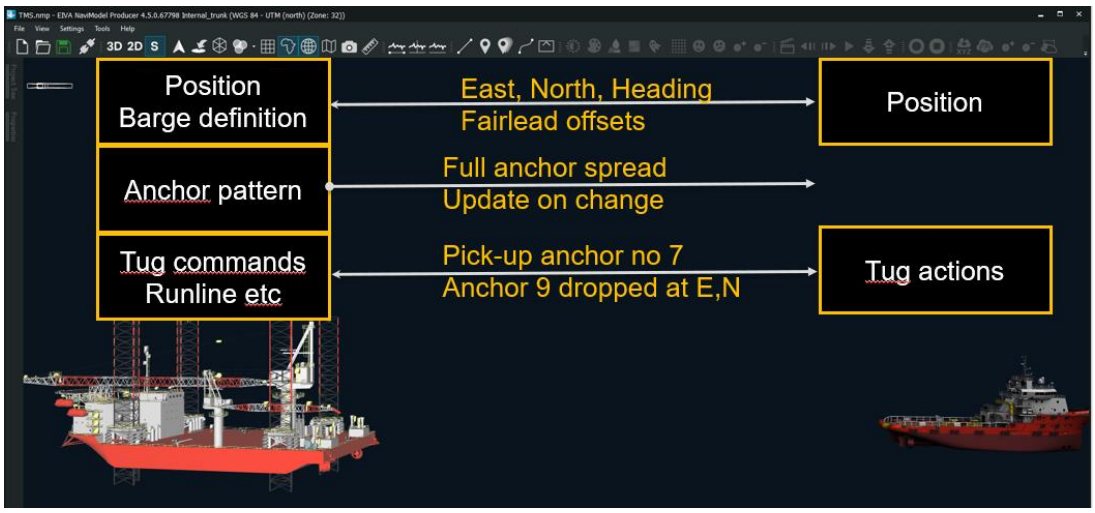


Figure 47 Sharing information between barge and tug(s)

To minimise the risc for collisions you might use data polling, where the bare (master) defines time slots for each vessels to send in

Please refer to dedicated documentation on data exchange and rig move operations

5.10 NaviPac eventing and triggering

NaviPac supports a series of events at kernel level – that is not to be compared with visual eventing from the Helmsman's Display

An event is defined by NaviPac generating an internal trigger, and calculating the position to that specific point of time. This can then result in special data outputs (eg message to NaviSuite trigger box, to air gun,), special data recording and listing on the Helmsman's Display

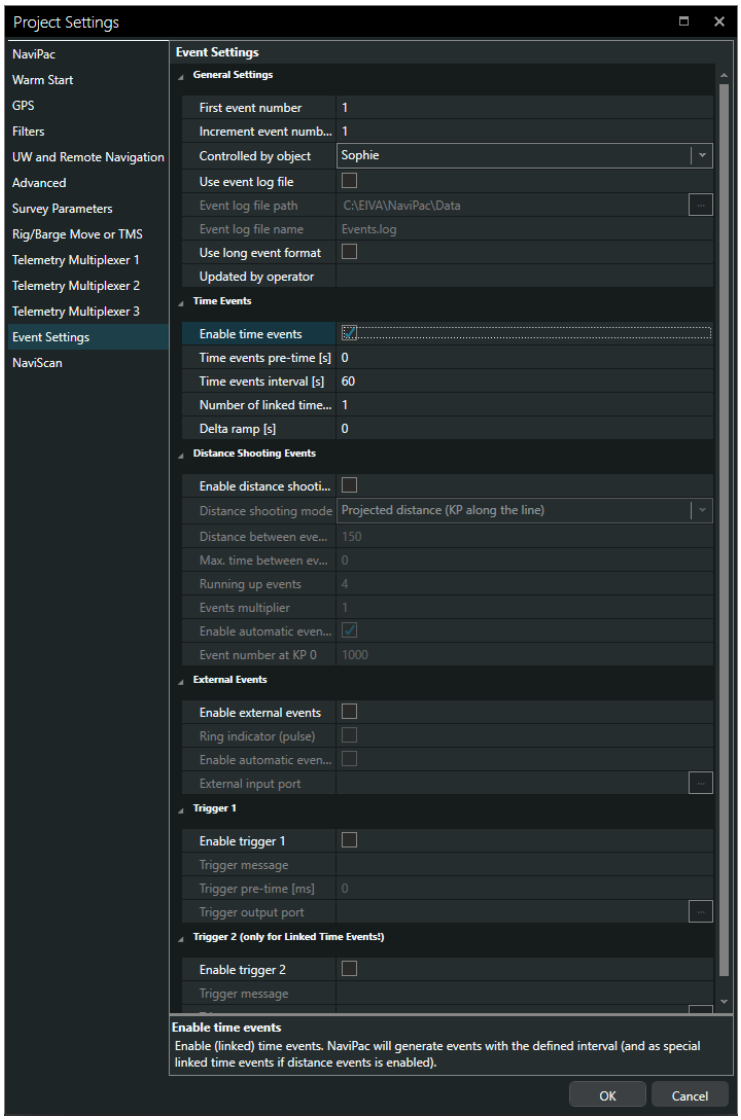


Figure 48 NaviPac event setting

- General setting
  Definition for event recording and global definitions
- Time events
  Time events, that is performing event with regular time interval. Requires active runline
- Distance events
  Events based on traveled distance – along line (KP) or just accumulated. Especially use oi n 2D seismic operations
- External events
  Input into NaviPac (serial port) creating the event

- Event trigger
  Output to port for triggering external device – typically used with trigger box

5.11 NaviScan

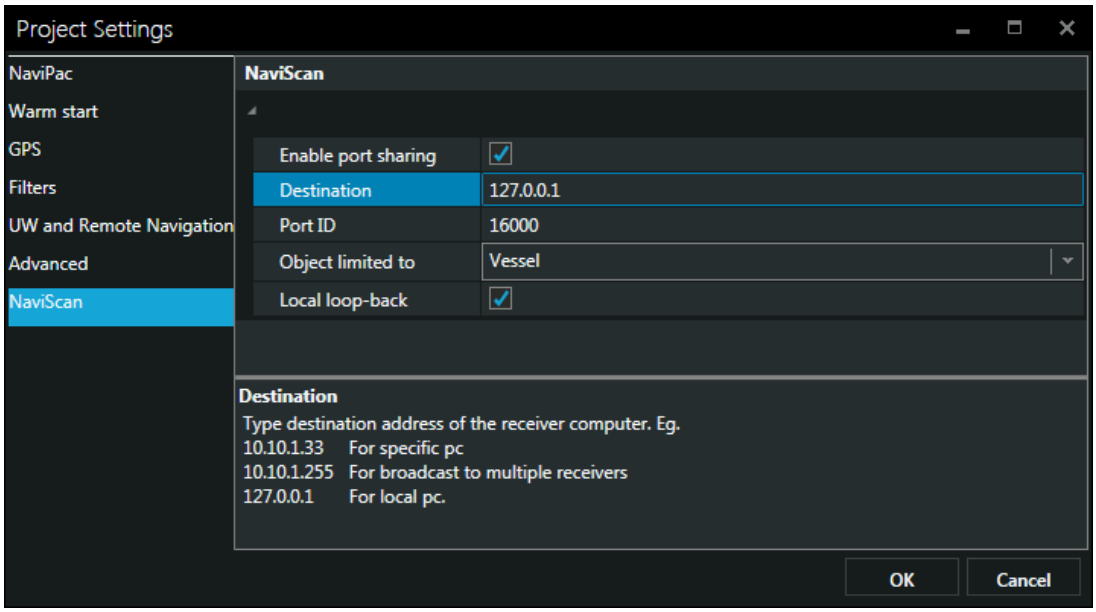


Figure 49 Sharing information with NaviScan utility

Enable port sharing
Define whether NaviPac should export data from data inputs to NaviScan and thus enable port sharing between the two programs without losing time-tagging accuracy. This is only applicable for serial (COM) inputs from:

- GPS
- Gyro
- Motion sensor
- Doppler log
- Bathymetry
- Pipe tracker

Sharing can be used on the same computer or on remote computer(s).

Destination
Define the destination TCP/IP address of the receiver computer. For example:

- 10.10.1.33 For specific computer receiving the data
- 10.10.1.255 For broadcast to multiple receivers on same subnet
- 127.0.0.1 For local computer only

Port ID
The data will be sent out as UDP/IP messages using the **TimeBox/ATTU** format. Each serial port will be output on its own UDP port, defined by the original port number plus this add-on. If you, for example, enter *10000*, then COM1 will be sent on UDP 10001.

Object limited to
Select whether or not you want NaviPac to broadcast all inputs (**NONE**) or only for a selected object (eg **ROV1**). The changes in setup will not be available until the next restart of navigation.

Local Loop Back
Enable you to send NaviScan data to both local IP and remote computer(s).

6 Navigation

The primary goal of NaviPac and **NaviPac Configuration** is to go online and collect data.

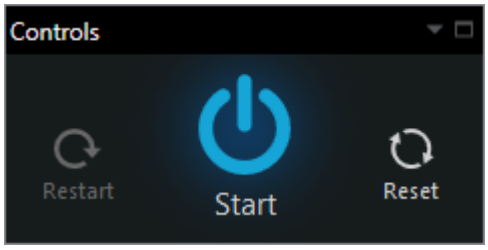


Figure 50 Navigation control toolbar

(See section 1.5 'Controls' of this manual for further details.)

When online navigation starts, NaviPac checks whether the setup file is compatible with the selections made in the online navigation. If so, NaviPac performs a warm start based on exactly the same online settings as the previous session. This is the preferable solution, as no special settings must be considered.

If the files are not compatible, perhaps due to addition or removal of instruments, NaviPac has to perform a cold (manual) start instead. This will result in two dialogue boxes opening.

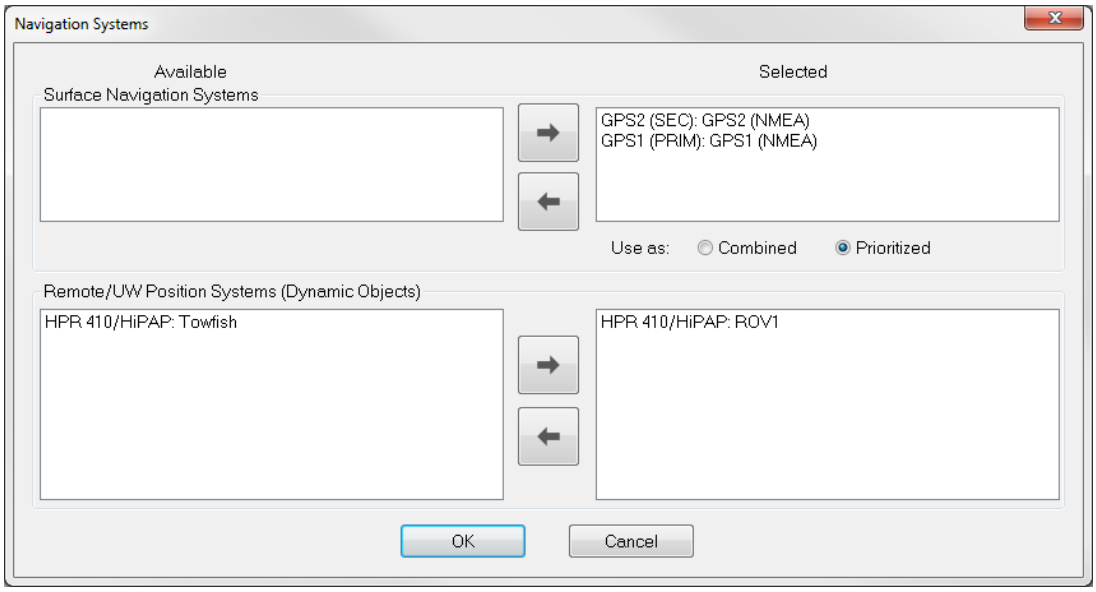


Figure 51 Online navigation manual start: Start by selecting instruments

Here, you must select at least one surface navigation input (eg the GPS1 shown in Figure 51). The selection can always be changed later on in NaviPac Online.

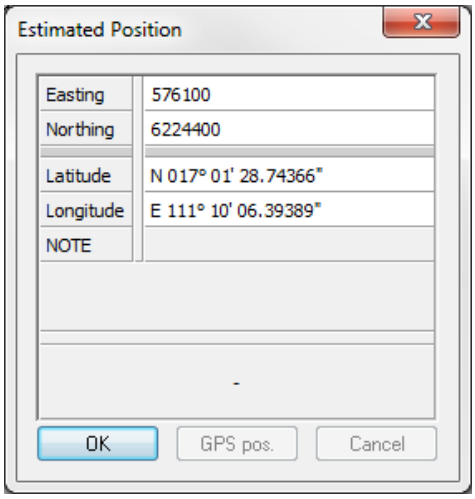


Figure 52 Online navigation manual start: Enter initial position for filter estimation

If you are using GPS, in the dialogue box shown in Figure 52, just click **OK**. If you use range/range or similar settings, you have to enter an approximate position to make the navigation operational.

7 References

You can find a total list of manual and documents on the EIVA documentation and FAQ site https://eiva.freshdesk.com/support/home (support.eiva.com)

8 Figures

Figure 1 NaviPac Configuration main window 4

Figure 2 Vehicle view illustration main vessel and ROTV 6

Figure 3 3D visualisation of vehicle and offsets 7

Figure 4 Define vehicle drawing and scale 8

Figure 5 NaviPac Project tree 9

Figure 6 Instrument properties for a gyro 11

Figure 7 Editing a number or text in a field 12

Figure 8 Selecting a file via browser function 12

Figure 9 Offset selection 12

Figure 10 Selection from a list 12

Figure 11 Using check boxes 13

Figure 12 I/O selection 13

Figure 13 NaviPac control buttons 13

Figure 14 NaviPac toolbar 14

Figure 15 Select instrument by port 14

Figure 16 Trace Settings view 16

Figure 17 NaviSuite License activator 17

Figure 18 About EIVA NaviPac 4 17

Figure 19 Edit instrument properties for a gyro 18

Figure 20 Instrument I/O dialogue box 21

Figure 21 Adding new instrument from top level 23

Figure 22 Adding new instrument from instrument group level 23

Figure 23 Selection of instrument group 24
Figure 24 Selection of the specific instrument type 25
Figure 25 Use of the search field 25
Figure 26 Finalising the instrument by editing its properties 26
Figure 27 Moving GPS2 in front of GPS1 27
Figure 28 Positioning a ROV via USBL 28
Figure 29 Display of current Geodesy setup 29
Figure 30 Selection of ellipsoid and projection type 30
Figure 31 Searching in EPSG database 32
Figure 32 Defining datum shift without EPSG 33
Figure 33 Defining datum shift when using EPSG 34
Figure 34 Definition of ITRF Reference Shift and Speed of change parameters 36
Figure 35 Apply ITRF Shift for an individual GPS instrument 37
Figure 36 Height reduction 38
Figure 37 Activating project setting 40
Figure 38 Parameters for NaviPac kernel 41
Figure 39 Selection of modules to include in Warm start 43
Figure 40 Global GPS definitions 44
Figure 41 Defining general filters 46
Figure 42 Global setting of remote positioning 48
Figure 43 Collection of miscellaneous settings 49
Figure 44 Defining survey parameters 51
Figure 45 Special setting for Rig move 52
Figure 46 Multiplexer configuration 53

Figure 47 Sharing information between barge and tug(s) 54

Figure 48 NaviPac event setting 55

Figure 49 Sharing information with NaviScan utility 56

Figure 50 Navigation control toolbar 58

Figure 51 Online navigation manual start: Start by selecting instruments 58

Figure 52 Online navigation manual start: Enter initial position for filter estimation .. 59